

Market-Wide Predictable Price Pressure[†]

By SAMUEL M. HARTZMARK AND DAVID H. SOLOMON*

We demonstrate that predictable uninformed cash flows forecast aggregate market stock returns. Buying pressure from dividend payments (announced weeks prior) predicts higher value-weighted market returns, with returns for the top quintile of payment days four times higher than the lowest. This holds internationally and increases when reinvestment is high and market liquidity is low. We estimate a market-level price multiplier of 1.9. These results suggest price pressure is a widespread result of flows, not an anomaly. (JEL G12, G14, G35)

Do trades in and of themselves generally move the level of the aggregate stock market? An answer to this seemingly simple question lies at the core of competing philosophies of how asset markets set prices. Almost all of asset pricing theory emphatically states no.¹ These theories posit that prices move because of information, and assume away any direct role for trade.²

Curiously, the generally skeptical attitude toward the importance of price pressure is not due to a lack of studies showing its existence. Trade has been shown to influence prices when companies are added to indexes (Shleifer 1986; Harris and Gurel 1986; Wurgler and Zhuravskaya 2002; Chang, Hong, and Liskovich 2015; Patel and Welch 2017), when mutual funds are forced to liquidate large positions of stocks (Coval and Stafford 2007; Lou 2012; Frazzini and Lamont 2008), and during the recent meme-stock episodes, such as GameStop (Bradley et al. 2021).

*Hartzmark: Boston College (email: samuel.hartzmark@bc.edu). Solomon: Boston College (email: david.solomon@bc.edu). Emi Nakamura was the coeditor for this article. We are grateful to Shaun Davies, Ken French, Christopher Jones, Ralph Koijen, Yueran Ma, Stefan Nagel, Lubos Pastor, Clemens Sialm, Eric So, Dimitri Vayanos, Michael Weber, and seminar participants at Amsterdam Business School, Bocconi University, Boston College, University of Central Florida, Chapman Experimental and Behavioral Finance Conference, Chicago Booth, Chinese University of Hong Kong in Shenzhen, City University Hong Kong, Curtin University, Deakin University, Econometric Society North American Winter Meeting, EDHEC, Hong Kong Polytechnic University, University of Houston, LeHigh University, University of Kentucky, Miami Behavioral Finance Conference, NBER Asset Pricing, NYU Stern, Singapore Management University, Shanghai Financial Forefront Symposium, University of Technology Sydney, Notre Dame Investment Management Conference, AIM Investment Conference, Democratize Quant Conference, and TU Munich for helpful suggestions. Some results were originally included in early drafts of Hartzmark and Solomon (2019), and an earlier version of this paper was titled “Predictable Price Pressure.”

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¹Or trades only move prices a very small amount. The field has moved past the basic efficient markets hypothesis intuition (e.g., that of Sharpe 1964; Ross 1976; and many others), but the general equilibrium models that follow yield similar intuition, as they predict economically negligible responses to price pressure. See Gabaix and Koijen (2021) for quantification of the demand curve slope in various models.

²Trade can only move prices in such models if it reveals information. The exact type of information varies based on the model. Prices move in standard rational asset pricing models due to information about cash flows or state variables related to discount rates, while in behavioral models information that shifts behavioral biases can influence prices, and in intermediary asset pricing models, it is news about intermediaries.

Instead, the concern seems to be that in trying to meet the identification burden (that is, finding trades not driven by information), the literature has mostly focused on individual stocks experiencing large and unusual flows in stretched circumstances. That is, while price pressure can exist in some cases, it is unclear whether such evidence is representative of the normal functioning of liquid markets, when flows are smaller, assets are deeper, and situations more predictable and recurring. Perhaps due to such concerns, assuming trade has no direct influence on price remains a part of almost every model of aggregate asset markets and is the stated opinion of most who study them.³

Provocatively, recent models (e.g., Vayanos and Vila 2021; Gabaix and Koijen 2021) argue that the standard assumption is misguided and that shifts in demand significantly influence price levels and dynamics of the overall stock market, regardless of whether such trades contain information. Yet providing the empirical evidence to justify such models encounters difficult identification problems. Gabaix and Koijen (2021) are forced to rely on structural estimates based on a novel technique requiring an entire stand-alone paper (Gabaix and Koijen 2024) to develop.

But even in these models of price pressure, the prediction is that prices mostly move because of information *about* a trade; that is, prices should move when markets learn that a trade is coming, not when the trade actually happens. For most trades, news of the trade occurs at the same time as the trade itself, so a price change cannot be separately ascribed to the news or the trade. However, if the trade were announced in advance, these models predict the price movement should occur largely on the news of the trade, with minimal movement when the trade occurs. If trades still significantly move prices, even when the information about the trade is fully known in advance, it suggests that the puzzle is considerably deeper than even current models of price pressure imply.

Thus, the seemingly simple question with which we began requires addressing a number of interesting and important questions. Is the evidence of trade moving prices of individual stocks in extreme settings relevant during normal times and at the market level? If trade moves markets, can clean and convincing evidence be provided with minimal structural assumptions? If trade does matter for market dynamics, is what matters largely information about trade, or the trade itself?

Our paper seeks to meet these challenges. We overcome the identification challenge by exploring a setting that allows for a direct reduced-form test of whether trades shift the level of the stock market in a normal setting that occurs almost every day. We are able to distinguish the direct effect of trades not only from any underlying economic information but also from news of the trade itself. We provide evidence for economically sizable price pressure (systematic shifts in uninformed demand moving prices) in a common and recurring setting for the overall stock market when all of the information about the flows is known weeks in advance.

We examine the impact of the predictable reinvestment of dividend payments in creating price pressure on the value-weighted market portfolio. Our identification

³Gabaix and Koijen (2021b) surveyed asset pricing academics as to their beliefs about the money multiplier (the price increase from each dollar of equity purchases) and found a median answer of zero, and among those with a positive answer, the median was an economically tiny 0.01.

is simple. When investors receive a dividend payment, some of the money is predictably invested into the market.⁴ While the actual purchases (which we do not observe) may contain information when they occur, all of the information necessary to understand the predictable component of flows from dividend reinvestment is revealed weeks prior on the announcement day. By using dividend payments as a proxy for the predictable component of flows due to dividend reinvestment, our analysis examines flows that are predictable in both timing and amount (as in Hartzmark and Solomon 2013) and occur almost every day. As a result, market participants should understand that these flows are uninformed (though we do not know if they actually do), and the recurring nature of the event gives liquidity providers the best opportunity to recognize the situation and counteract it.

We find that when daily dividend payouts are higher, daily value-weighted market returns are significantly higher. The magnitude of this effect is considerable. A 1 standard deviation increase in payout is associated with higher returns of 3.2 basis points (compared to an unconditional mean return of 4.1 basis points). Value-weighted market returns are four times higher in the highest quintile of dividend payment days than in the lowest. Days with an above-median dividend payment account for 78 percent of the equity premium.

This is not due to shifts in the denominator of stock market capitalization. Similar effects are obtained with year-by-month fixed effects or with a measure of abnormal dividends based on the difference from the longer-term average daily dividend. These effects survive controlling for day-of-the-week effects, turn-of-the-month effects (e.g., Lakonishok and Smidt 1988), Fed announcements (Lucca and Moench 2015), and other macroeconomic announcements (Savor and Wilson 2013). We next show that this result holds on average in 58 international markets, which allows us to control for worldwide daily patterns in returns. If a country has higher dividends on a given day relative to other countries, it tends to have higher returns relative to other countries on that day.

We also show that predictable dividend payments form the basis of a potentially profitable trading strategy, at least before costs. Each strategy increases market exposure (by increasing leverage) on days with high dividend payments and decreases market exposure on days with low dividend payments. While the market earned annualized excess returns of 7.2 percent per year, our strategies produce annualized returns from 8.7 percent (with a maximum leverage of 1.5) to 14.0 percent (with a maximum leverage of 3). Each strategy has a higher Sharpe ratio than the market but takes on approximately the same amount of market risk as the market portfolio, having average leverage of about one and a market beta of about one. The alphas compared to standard risk models of our maximum 3x leverage strategy are approximately 7 percent per year, which is similar in magnitude to the equity premium itself. While the strategy has high transaction costs due to high turnover, it operates largely independent of, and thus can be layered on top of, other strategies.

Consistent with price pressure from uninformed demand, market predictability is related not just to the amount of dividends but also their likelihood of reinvestment. Asset managers typically make their largest payouts in the fourth quarter, making

⁴Though clearly not all. For example, some dividends are consumed (e.g., Hartzmark and Solomon 2019; Di Maggio, Kermani, and Majlesi 2020; Baker, Nagel, and Wurgler 2006; Bräuer, Hackethal, and Hanspal 2022).

them less likely to reinvest dividends at such times (as they need to shortly pay out the cash to their own investors). During the fourth quarter, dividend payments do not predict market returns. During the first quarter, when dividend investment is likely highest (as managers have a long time before needing to pay out again), we find the largest effect. A 1 standard deviation increase in dividend payments in the first quarter is associated with an increase in market returns of 6.4 basis points, and the alpha to our maximum 3x levered trading strategy of 6.5 basis points per day is more than double the equity premium. Evidence from the time series is also consistent with the importance of asset managers. The impact of dividend price pressure has increased since roughly 1990, as mutual funds, ETFs, and other asset managers have become a larger component of equity holdings. Consistent with liquidity being important for price pressure, we find larger effects when proxies of market liquidity are lower.

Up to this point, we have been agnostic over whether the price impact is permanent or temporary. Examining cumulative returns, we find no evidence of a reversal over the month after dividend payment. If anything, the data suggest a continuation, consistent with some investors reinvesting dividends with a delay. Extending out further adds significant noise to our estimates, and one quarter after payment we cannot reject either a permanent effect or a full reversal. While our setting offers a number of advantages, it does not allow precise quantification of the total extent and the timing of a reversal in the time series.

Our baseline strategy uses stale information and predicted purchases, rather than actual purchases, to ensure that payment amounts do not have any information released on the payment date itself. A potential concern is that our results reflect a delayed reaction to the information contained in the announced dividends. To demonstrate this is unlikely to account for our results, we approximate a day's dividend payment using dividend payment from the same calendar day in prior years. After showing this rule explains a significant portion of the variation in daily dividend payment, we show that much of the returns can be predicted using data as old as 10 years prior. Thus, the returns are unlikely to reflect a delayed reaction to news from recent dividend announcements.

The flows we examine are predictable and recurring, which means that a rational arbitrageur *should* recognize them as uninformed, but we do not know if they actually do so. If they do, it is puzzling under standard asset pricing models why prices move at all, and it is unclear what the friction is that means liquidity providers are unable to offset such flows. Under models of price pressure, it is unclear why prices shift on the payment date as opposed to when the news about the flows is released. If the market does not identify these flows as uninformed, then this misperception can account for the effect. The space of flows liquidity providers are meant to offset is large, and this is one of the more straightforward and manageable to be aware of. If these flows are missed, then it suggests the ability of liquidity providers to identify and react to uninformed flows is much more limited than commonly assumed. A definitive answer as to which scenario best describes our setting is beyond our paper, but our results are novel under each set of assumptions.

Finally, we perform a back-of-the-envelope calculation to estimate the multiplier of prices with respect to reinvested dividends, similar to Gabaix and Koijen (2021). This represents the magnitude of a market capitalization increase for a dollar of dividends reinvested (i.e., the inverse of elasticity). Our baseline estimate of 0.67

represents a lower bound on the multiplier (assuming 100 percent reinvestment) and demonstrates that standard models and survey respondents greatly underestimate its magnitude. Estimating reinvestment rates for different investor groups and combining this with our results gives a multiplier of 1.9. This is somewhat lower than that in Gabaix and Koijen (2021), showing potentially that the size of the multiplier estimated may be sensitive to methodology and which flows are used for estimation. Nonetheless, our results are puzzlingly large under their model. In their framework, prices move when information about a trade is announced. While this normally occurs at the time of the trade itself, the flows we study are all announced in advance and so ought to have only a minimal effect on prices.

This paper fits into several strands of literature showing the existence of price pressure. A number of papers have explored reactions to one-off events in the cross section. When companies are added to an index, this results in an increase in price (Shleifer 1986; Harris and Gurel 1986; Wurgler and Zhuravskaya 2002; Chang, Hong, and Liskovich 2015; Patel and Welch 2017). A related literature on fire sales explores how flows in mutual funds can induce price pressure from predictable purchases and sales of underlying assets (Coval and Stafford 2007; Lou 2012; Frazzini and Lamont 2008). Ben-David et al. (2020) argue that flows from Morningstar ratings changes induce price pressure to individual stocks and influenced the dynamics of cross-sectional style returns. Various papers have analyzed stock-level effects of dividend payouts. Hartzmark and Solomon (2013) show price pressure from dividend-seeking investors in the lead-up to a dividend ex-day and negative returns thereafter, an event that is also both predictable and recurring. Berkman and Koch (2017) show positive payment date returns for dividend-paying stocks that have dividend reinvestment plans. Other papers show cross-sectional patterns where stocks owned by mutual funds receive spillover price pressure when the funds receive a dividend or a merger payment (Schmickler 2021; Chen 2024; Kvamvold and Lindset 2018). Most of this evidence posits effects through overlapping holdings (Chen 2024; Kvamvold and Lindset 2018) and/or industries (Schmickler 2021).⁵ While these cross-sectional patterns raise the question of whether there may be aggregate time series patterns, the existence of such a pattern is far from obvious from these results.⁶

Evidence of price pressure for *entire markets* is harder to find. While this paper focuses on equity markets, other papers have explored the influence of supply and demand using bonds (Vayanos and Vila 2021; Greenwood and Vayanos 2014; Greenwood and Hanson 2013; Lou, Yan, and Zhang 2013; D'Amico and King 2013), mortgage backed securities, (Gabaix, Krishnamurthy, and Vigneron 2007), and options (Garleanu, Pedersen, and Poteshman 2008). Closest to our paper in the examination of equity markets is Gabaix and Koijen (2021), who use granular instrumental variables to identify shocks to institutions to estimate the multiplier of flows on prices. Parker, Schoar, and Sun (2020) show mechanical rebalancing

⁵ Kvamvold and Lindset (2018) perhaps come the closest to our results. While they show daily price pressure in stocks connected by mutual fund holdings, they show inconsistent effects in S&P 500 stocks. Given their emphasis on cross-sectional predictions based on mutual funds minimizing tracking error, they examine a short time period and make a number of specification choices that make it not obvious that the value-weighted market return is predicted by daily dividend payouts.

⁶ Cross-sectional predictors are not generally good predictors of time series returns (Engelberg et al. 2023).

by target date funds influences the cross section of stock returns and overall market dynamics. Da et al. (2018) finds market-wide price pressure in Chile for stocks and bonds following recommendations for pension allocations, and Li, Pearson, and Zhang (2020) show that flows based on IPO regulations influence the Chinese market. Evidence on the effects of predictable flows at the market level has not, to our knowledge, been studied. Our paper complements this work by providing evidence of predictable price pressure through a simpler identification channel where the argument for the flows being uninformed is strong, where the flows are predictable in both timing and amount, and the event is a normal day-to-day occurrence.

Our paper contributes to the literature on the predictability of market returns. The long-standing conclusion of this literature (using variables like past returns or price/dividend ratios) is that while there is considerable predictability at long horizons of several years, there is minimal predictability at horizons less than a year (Fama and French 1988; Campbell and Shiller 1988; Campbell and Shiller 2001).⁷ In contrast, we document significant predictability at short horizons, and associated trading strategies that are implementable and profitable (before expenses). Notably, this comes from variables that themselves vary over short horizons and are quite different to the slower-moving variables generally studied. The closest related finding is the FOMC announcement effect in Savor and Wilson (2013) and Lucca and Moench (2015) and calendar effects in Lakonishok and Smidt (1988).⁸ It is striking, however, that this very small group of papers represent the short horizon predictors of market returns that need controlling for (and they do not explain our effect). This contrasts sharply to the number of longer-term predictors of market returns, and even more so to the cross section of stock returns, where academics cannot agree to the nearest hundred how many anomalies need to be controlled for.⁹

Our result is that the expected amount of trading (based on information announced far in advance) moves prices on the day of the trade itself, which is difficult to explain under existing price pressure models (e.g., Vayanos and Vila 2021; Gabaix and Koijen 2021). In a sense, it is an almost mechanical prediction that sufficiently large liquidity-demanding orders will move prices as long as liquidity is finite. What is surprising is that liquidity providers, even with advanced notice, are unwilling or unable to offset such effects at the time of trade. The reasons why this is are of considerable interest for future study.

I. Empirical Strategy

Our key testable prediction relative to the textbook framework is that predictable uninformed flows will shift prices. In order to test this prediction, a measure of flows is needed that should be considered uninformed under standard theories. Such flows

⁷Even at long horizons, Welch and Goyal (2008) argue that much of the predictability disappears out of sample and thus is unlikely to be useful for trading. Meanwhile, recent research, such as Farmer, Schmidt, and Timmermann (2023), fails to find much short-term predictability from a variety of variables, other than during short periods of time.

⁸Another daily market-wide return predictor is variation in sunlight (e.g., Kamstra, Kramer, and Levi 2003; Hirshleifer and Shumway 2003), although this seems unlikely to be related to dividend payout amounts.

⁹McLean and Pontiff (2016) examine 97 anomalies; Harvey, Liu, and Zhu (2016) study 313; Chen and Zimmermann (2022) study 319; Hou, Xue, and Zhang (2020) study 452.

ought to not only *be* uninformed but should also be easily *recognized* as being uninformed by a rational investor.

One way to do so is to focus on the component of flows that are *ex ante* predictable long before actual trading occurs. The rationale is analogous to why asset pricing tests use information known prior to an event as a measure of *ex ante* predictable returns. *Ex post* realized flows likely contain information that could not be known before the trade occurs. By conditioning on the *ex ante* component of information, we do not include the total information in actual flows that could only be known only *ex post*. In doing so, we are conditioning on the information the market had before trading occurred, making it a good proxy for the predictable component of flows that a textbook market would understand to be uninformed.

To illustrate the intuition using our dividend setting, assume there is an announced dividend that will be paid on a future date t . It is likely that the difference between predicted flows (based on the information from the dividend announcement) and *ex post* realized flows occurring on date t contains information that shifts prices. For example, assume that when the payment is made, more money is reinvested than would have been expected at the time of the announcement. This behavior could indicate positive news, which in a rational market could lead to a price increase of the market. Similarly, assume the payout is used to purchase certain stocks and other stocks are not purchased. This could release positive news for the purchased stocks and negative news for those that were not, leading to cross-sectional differences in returns.

The difference in prediction between our framework and the textbook framework comes from the influence of predictable flows, not realized flows. A test focusing on a realized flows could indeed reflect news in a rational market that was unknowable before the trades occurred. By using only predictable flows based on information released long before trades occur, the flows cannot reflect news released when trading finally happens and thus in any textbook framework are not predicted to meaningfully influence prices upon payment.

II. Institutional Details about Dividend Payments

Our tests examine the impact of aggregate dividend payments on stock market returns. Aggregate dividend payments represent an interesting source of uninformed trade because their timing is completely predictable well in advance of the date the payments actually occur. The fact that the payments are known in advance, and result in predictable reinvestment, offers clean identification of whether market prices are impacted by uninformed flows. This section discusses the relevant institutional details of dividend payments as they relate to the plausibility of our identification.

There are three relevant dates that occur for a normal cash dividend.¹⁰ The first is the announcement date, on which companies declare the timing and amount of a dividend. The next date is the *ex-dividend* date, the first day when those who buy

¹⁰Our study focuses on ordinary cash dividends (Center for Research in Securities Prices distid starting with “12”) and third digit code of 0 to 5. These are the most commonly occurring and are standardized in the manner and timing of how they are paid.

the share will not receive the dividend. The payment date is the date when cash is actually disbursed.

Due to the logistics necessary to accurately send dividend payments to the correct investors, these events occur with a significant lag between each of them. Payment occurs on average 43 days after the dividend announcement and 22 days after the ex-dividend date. While there is uncertainty in the timing and amount of dividends that companies will declare, this is resolved when the dividend is announced. The ex-date is when any implications of receiving a dividend payment, such as tax consequences, are resolved. We focus on the payment date, which is the date when cash is disbursed and lacks economically meaningful news or tax implications.

A potential concern is whether there is meaningful economic uncertainty that is resolved when a payment is actually made. For example, assume that there is systematic uncertainty for the overall stock market the day before a dividend is paid that is resolved the next day upon payment. Further assume that this uncertainty is correlated to the level of the payment yield. In such a case the return from market close the day before a payment date (where the price would reflect this uncertainty) to market close on the payment date (where the price would no longer reflect this uncertainty) would be increasing in the level of the dividend payment, which is the empirical pattern that we find. As we discuss below, such an explanation is unlikely to account for our results.

In practice, if a payment of a given dividend is imminent, it has always been paid. We sought to find any example where a normal cash dividend was declared, went ex-dividend, and was not paid. Despite trying hard to find such an example, we were unable to do so. We also used ChatGPT with a plug-in to search SEC filings, which also was unable to come up with any example.¹¹

We note that this is consistent with the regulatory structure in which dividends are declared in the United States. Once the dividend is declared, companies have a legal obligation to pay it.¹² Perhaps more importantly, there are significant restrictions, which mean that only firms in a strong financial situation are able to declare a dividend in the first place.¹³ In the Supplemental Appendix we explore a variety of measures of firm health and find that such laws seem to have effectively made dividend-paying firms significantly stronger than non-dividend-paying firms based on accounting measures.¹⁴

Further, the duration of returns that we study is extremely short. In order for news to drive our results, it would need to occur during the 24 hours between the market

¹¹ Using ChatGPT 4 with the AITickerChat plug-in (to search SEC filings) on August 12, 2023, we asked it to find instances of “firms that announced a cash dividend, the ex-dividend date passed and the stock went ex-dividend, but the dividend was not paid on the announced payment date.” The reply was, “To my knowledge, as of my last training cut-off in September 2021, there hasn’t been a prominent case in the United States where a major publicly-traded company has failed to pay a declared dividend after the ex-dividend date has passed. There might be instances in smaller markets or with smaller companies, but they would not be well-documented or widely reported.”

¹² Even in the event of corporate bankruptcy before the payment date, shareholders are unsecured creditors for the amount of the dividend.

¹³ While there is some variation across state laws, generally, firms may only pay dividends if they are currently running a surplus or if they have achieved net profits in the current and former fiscal year. See § 170 of the Delaware General Corporation for an example: <https://delcode.delaware.gov/title8/c001/sc05/index.html>.

¹⁴ For example, while about 60 percent of non-dividend-paying firms are profitable, more than 90 percent of dividend-paying firms are (t -statistic of 72.35). Supplemental Appendix Table IA.V shows similar large differences when examining z -score of financial distress, earnings to assets, and return on assets (ROA).

close the day before and the market close on the payment date. If for some reason the payment were canceled earlier, investors ought not to respond to this on the payment date. Given we were not able to find examples where a dividend was not paid in the longer period between the ex-date and payment, we clearly could not find any example of a dividend being canceled after hours the night before the payment date.

We note that the lack of meaningful resolution of uncertainty, or any other plausible economic meaning, is reflected by the academic literature. While there are large literatures exploring information released on announcement dates and large literatures examining the tax implications for returns on the ex-date, there are no papers that we are aware of that propose a resolution of uncertainty (whether rational or driven by a behavioral bias) occurring on the payment date.

Finally, our results hold when we measure market returns that exclude firms that actually pay a dividend, so the resolved uncertainty must be relevant to the aggregate economy. A profitable firm being unable to pay a dividend after hours the night before payment has never (to our knowledge) occurred, but if it were to, it seems that the only plausible mechanism would be something like an extreme fraud that was uncovered in this brief time period. But such a mechanism is likely to be highly idiosyncratic, and thus, there would be no reason why it would move the entire stock market.

As a consequence, dividend payment amounts are predictable well before they occur. Market-wide dividend payments also exhibit substantial day-to-day variation due to idiosyncratic differences between individual companies' payment dates as well as large differences in firm size. Further, while there is seasonality in dividend payments, about 90 percent of trading days involve a dividend payment.

An increase in cash must end up somewhere; it can be consumed, left in cash or the money market, or invested into risky securities. We focus on the latter as a potential driver of returns. Hartzmark and Solomon (2019) show that investors of any type, including institutional investors and mutual funds, do not generally reinvest dividends into the securities from which they came. Given the paucity of same-stock reinvestment, this raises the question of whether the remainder of the dividend amounts are invested into other stocks, which may be inducing price pressure. To consider the broadest and most liquid set of test assets, we take as our main dependent variable the value-weighted market portfolio from Center for Research in Securities Prices (CRSP).

The final question is the timeline under which people use a dividend payment to purchase securities. Dividend reinvestment likely occurs with different timing for institutional reasons, differences in attention, and differences in strategy. If investors trade once their cash balances update, differences in reinvestment timing could be due to differences in when the cash appears in an account. For example, some institutions update balances throughout the day (allowing for trade on $t = 0$ and after), though many banks clear each day's deposits that night (allowing for trade on $t = 1$ and after).¹⁵ Further, not every attentive investor has a strategy that trades every day. In addition, some investors may not be attentive to a dividend payment and reinvest only when the higher cash balance comes to their attention. We lack details for the

¹⁵ There is likely heterogeneity across institutions as to when in a day a payment appears in an account, even if the payment appears on the same day. Further, we expect that in the early period of our sample, there was likely more heterogeneity and longer lags for a payment to clear, though we lack documentation from this period.

myriad of institutions, time periods, and strategies involved, so while we predict there will be reinvestment on the payment date and the days immediately following it, we are agnostic as to which specific day should have the largest effect.

III. Data

Stock return data are from the CRSP, including returns, dividend amounts, and payment dates. We limit our return data to ordinary common shares (codes 10 or 11) listed on the NYSE, NASDAQ, or AMEX (CRSP n.d.b). For dividend amounts, we examine ordinary cash dividends. We examine data from 1926 through 2018. Monthly mutual fund holdings come from CRSP (CRSP n.d.a). International stock returns data come from Compustat Global (Compustat n.d.). The Supplemental Appendix contains further discussion of the data and variable construction.

Unless otherwise noted, when we refer to the timing of a dividend, we mean the payment date. For example, when we refer to a dividend date or a dividend yield, we are referring to date or yield based on the payment date. The major exception to this is when we examine mutual fund dividends, as there is not payment date information for fund dividends, so the ex-date is used.

IV. Results

A. Aggregate Dividend Payments and Market Returns

Table 1 examines the effects of daily payment yield on value-weighted market returns. The independent variable is the dividend payment yield on a payment day t , which is the daily total dividend payments, divided by the previous day's total market capitalization. If a dividend payment is invested on the payment date and induces price pressure, the market return on date t would be predicted by the dividend payment on date t , while if it is invested the day after the payment date, the market return on date t would be predicted by the dividend payment on date $t - 1$. In column 1, we consider dividend payments at various lags from zero days to four days. This shows a coefficient on payment yield of 55.76 for day t (t -statistic of 1.74), 60.04 for day $t - 1$ (t -statistic of 2.73). Days $t - 2$ and $t - 3$ are smaller and statistically insignificant, though day $t - 4$ is also larger and marginally significant (t -statistic of 1.89). In terms of magnitudes, a 1 standard deviation increase in payout yield (0.0003176) predicts higher returns of 1.8 basis points on day t and 1.9 basis points on day $t + 1$.

Payout yield uses the market level as a denominator, so we wish to make sure that the effects capture variation in dividends, not variation in the market level. We deal with this problem in several ways. As a first step, in column 2 we include a year-by-month fixed effect. If the concern is that the price-to-dividend ratio predicts long-term periods when the market has a higher or lower return, a year-by-month fixed effect will control for general differences in such an average. This also controls for specific calendar effects (e.g., the January effect in Rozeff and Kinney 1976 and Thaler 1987). The inclusion of this fixed effect increases the magnitude and significance of the coefficients, to 74.85 on day t (t -statistic of 2.26), 71.98 on day $t - 1$ (t -statistic of 3.10), and 66.66 (2.32) on $t - 4$ (with days $t - 2$ and $t - 3$ also being slightly larger, though insignificant).

TABLE 1—MARKET RETURNS BASED ON DIVIDEND PAYMENT

	Value weighted						
	(1)	(2)	(3)	(4)	(5)	(6)	
<i>Panel A. Baseline results</i>							
Mkt div pay[$t - 1, t$]			59.50 (3.32)	67.07 (3.47)			
Mkt div pay[t]	55.76 (1.74)	74.85 (2.26)					
Mkt div pay[$t - 1$]	60.04 (2.73)	71.98 (3.10)					
Mkt div pay[$t - 2$]	23.59 (0.98)	35.98 (1.43)					
Mkt div pay[$t - 3$]	14.56 (0.50)	25.73 (0.85)					
Mkt div pay[$t - 4$]	53.96 (1.89)	66.66 (2.32)					
Mkt div pay[$t + 1$]					14.57 (0.59)	26.48 (0.99)	
Mkt div pay[$t + 2$]					-25.23 (-1.07)	-12.73 (-0.51)	
YM fixed effects	No	Yes	No	Yes	No	Yes	
R^2	0.00105	0.0503	0.000700	0.0498	0.0000697	0.0491	
Observations	24,534	24,534	24,537	24,537	24,536	24,536	
	SP 500 (1)	Equal weighted (2)	Quint 1 (3)	Quint 2 (4)	Quint 3 (5)	Quint 4 (6)	Quint 5 (7)
<i>Panel B. Alternative portfolios</i>							
Mkt div pay[$t - 1, t$]	70.04 (3.26)	72.35 (3.50)	94.01 (3.88)	69.74 (3.09)	64.95 (3.07)	63.74 (3.01)	62.32 (3.24)
YM fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0396	0.0858	0.114	0.0925	0.0756	0.0648	0.0518
Observations	14,221	24,537	24,537	24,537	24,537	24,537	24,537

Notes: This table shows how the market return varies with the dividend payment yield. In panel A, market return is measured in percent as the CRSP value-weighted market index. Market dividend payment yield is measured as the cumulative payment yield on day t and $t - 1$ (Market Div Pay[$t - 1, t$]), or the payment yield on the indicated day scaled by total market cap from the prior day. Even-numbered columns in panel A contain year-by-month fixed effects. Panel B repeats the regressions using different portfolio returns. It examines the S&P 500, CRSP equal-weighted market index, and quintiles of firm size where 1 is the smallest and 5 the largest. All columns in panel B contain year-by-month fixed effects. t -statistics based on heteroskedasticity-robust standard errors are in parentheses.

Given this daily pattern, in Table 1, columns 3 and 4, we focus on the main variable we will use, the cumulative dividend payment yield on the payment date and the day before.¹⁶ Column 3 finds similar effects to column 1; dividend payout from day t and day $t - 1$ positively and significantly predicts value-weighted market returns on day t (coefficient of 59.50 and t -statistic of 3.32). When year-by-month

¹⁶We find the $t - 4$ effect likely reflects delays in receiving the dividend in the early period of our sample when checks were all physically mailed out, so we mostly focus on the first two days where the predictions are clear. Running the column 2 specification using data prior to 1960 yields a coefficient on Mkt Div Pay[$t - 4$] of 79.9 (t -statistic of 1.83); post-1960 this coefficient is 45.73 (t -statistic of 1.58), and post-1980 is 32.23 (t -statistic of 0.73).

fixed effects are added in column 4, the results are again stronger, with coefficients increased to 67.07 and a t -statistic of 3.47. A 1 standard deviation increase in payout yield over days t and $t - 1$ (0.0004711) predicts higher daily returns of 3.2 basis points on day t .¹⁷

Table 1, panel B uses different measures of market returns. Column 1 uses the S&P 500, perhaps the most likely instrument to be traded, and finds a similar effect to the baseline. Column 2 examines an equal-weighted portfolio, and columns 3 through 7 explore portfolios based on quintiles of size. Moving from small to large, there are decreasing effects, consistent with the predictions of price pressure, assuming that size is a proxy for liquidity. Most importantly, no matter what portfolio is examined, we observe economically and statistically strong effects on returns from predictable flows.

If the returns reflect price pressure from investing a dividend payment, then there is less reason to expect positive return predictability before the payment is made.¹⁸ Thus, the period directly prior to dividend payment offers a placebo where price pressure does not make strong predictions about returns. To examine this period, Table 1, panel A, columns 5 and 6, examines the influence of future dividends (on days $t + 1$ and $t + 2$) on current (day t) returns. The coefficients are economically small and insignificant. This is consistent with dividend payments only influencing returns after they occur. Incidentally, this also suggests that there is not price pressure on these days coming from investors front-running the effects we document.

While year-by-month fixed effects remove slow-moving trends in the market price, it is possible that the effects are driven by day-to-day fluctuations in the denominator. We employ a number of empirical strategies to ensure that this is not the case. In the Supplemental Appendix we show similar results scaling the dividend payment amount by market value at various different lags, rather than the prior day's value.

While day-to-day fluctuations in the price cannot explain the pattern, there may be an unaddressed concern with using price in any manner. To address such a concern, in Table 2, panel A, we examine a measure of dividend payment that does not employ market prices at all. Specifically, we construct an abnormal dividend yield, which is dividends on days t and $t - 1$ divided by the average daily dividend paid over the prior year.¹⁹ When market returns are regressed on abnormal dividends, the results are positive and significant at the 1 percent level, when returns are value weighted (columns 1 and 2) or equal weighted (columns 3 and 4), and regardless of whether year-by-month fixed effects are included. Taking the value-weighted estimate in column 2, a 1 standard deviation increase in the dividend yield is associated with an increase in expected returns of 2.7 basis points, similar to the 3.2 basis points found from the same specification normalizing by market capitalization.

¹⁷ Mkt Div Pay[$t - 1, t$] has a mean of 0.00025 and a standard deviation of 0.0004711.

¹⁸ In principle, an attentive investor could trade before the payment date since cash is only required at settlement. The settlement date occurs two or three days after the purchase date, depending on the sample period, so one could buy earlier and have the cash arrive before settlement. An investor with access to leverage could borrow against a future dividend and trade as soon as it was announced, though they would need a process in place to trigger a trade at this earlier date.

¹⁹ We calculate this average using the trading days from $t - 20$ to $t - 272$. The average year contains 252 trading days, and we skip about a month (20 trading days) to ensure the denominator excludes any recent market information.

TABLE 2—MARKET RETURNS BASED ON DIVIDEND PAYMENT: ALTERNATIVE DIVIDEND NORMALIZATIONS

	Value weighted		Equal weighted					
	(1)	(2)	(3)	(4)				
<i>Panel A. Abnormal dividend</i>								
Mkt abnormal div[$t - 1, t$]	0.00698 (3.39)	0.00786 (3.58)	0.00637 (2.89)	0.00790 (3.43)				
YM fixed effects	No	Yes	No	Yes				
R^2	0.000499	0.0497	0.000424	0.0857				
Observations	24,266	24,266	24,266	24,266				
	2 weeks		Quarter		Third		Median	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel B. Top dividend payment days</i>								
Top days	0.0983 (2.39)	0.116 (2.75)	0.0568 (3.56)	0.0709 (4.27)	0.0443 (3.04)	0.0573 (3.71)	0.0311 (2.28)	0.0473 (3.15)
Constant	0.0370 (5.36)	0.0363 (5.22)	0.0266 (3.41)	0.0231 (2.92)	0.0260 (3.15)	0.0217 (2.57)	0.0252 (2.63)	0.0171 (1.69)
YM fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
R^2	0.000315	0.0495	0.000535	0.0498	0.000387	0.0497	0.000215	0.0495
Observations	24,287	24,287	24,286	24,286	24,286	24,286	24,286	24,286

Notes: This table shows how the market return varies with the dividend payment yield using different dividend normalizations. In panel A, the dividend payment yield is measured as the sum of dividends paid on days t and $t - 1$ divided by the average daily dividend payment on days $t - 20$ through $t - 272$. In panel B, “Top days” shows the coefficient on a dummy variable if the dummy variable listed in the column header is equal to 1. The “2 weeks” columns include a dummy variable equal to 1 if the payment yield is in the top 10 of the prior 252 trading days, “Quarter” include a dummy variable equal to 1 if the payment yield is in the top 63 of the prior 252 trading days, “Third” include a dummy variable equal to 1 if the payment yield is in the top 84 of the prior 252 trading days, and “Median” include a dummy variable equal to 1 if the payment yield is in the top 126 of the prior 252 trading days. Even-numbered columns contain year-by-month fixed effects. t -statistics based on heteroskedasticity-robust standard errors are in parentheses.

Figure 1 examines how value-weighted returns vary with this abnormal dividend.²⁰ We split days into quintiles based on the abnormal dividend and take the average of the value-weighted market returns in each quintile. The graph shows a monotonically increasing relationship between payment yield and returns. The returns in the lowest two quintiles are about 2 basis points. For the next two quintiles, this value increases to roughly 4 basis points. The top quintile of abnormal dividends is associated with value-weighted market returns of about 8 basis points.

The graph suggests that the largest effect is concentrated in more extreme dividend payment dates. To test this, and to use a simple rule that is easy to interpret and ex ante tradable, we examine whether or not a given day's dividend payment is high relative to the payments made in the previous 252 days. Specifically, we regress market returns on dummy variables equal to 1 if the cumulative dividend payment on day t and $t - 1$ is in the top 2 weeks (10 days), quarter (63 days), third (84 days), or half (126 days) of days in the past year, and equal to 0 otherwise. These regressions display the magnitude of the return effect from the regression coefficient, as

²⁰In addition to not being influenced by daily variation in the market price, the abnormal dividend is also largely independent of long-run changes to the market dividend yield (e.g., Fama and French 2001).

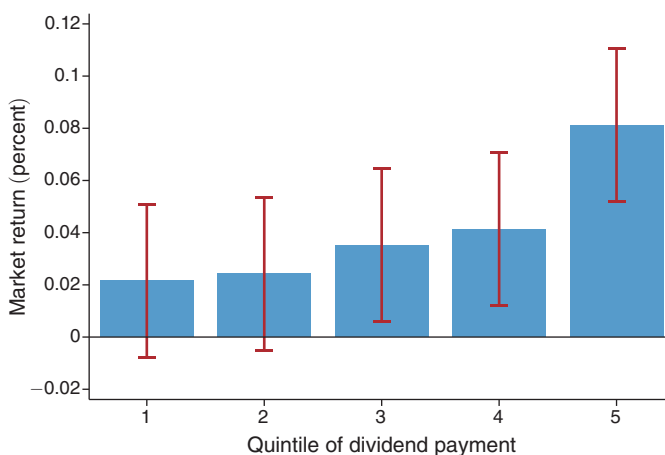


FIGURE 1. DAILY MARKET RETURNS BASED ON DIVIDEND PAYMENTS

Notes: This graph shows the daily value-weighted market return (in percent) on day t based on the quintile of the abnormal dividend payment on day t and $t - 1$. Red bars represent 95 percent confidence intervals.

it represents how different the return on these extreme dividend payment days is relative to the other days.

Table 2, panel B, column 1 shows that with no controls, the top 10 days have higher value-weighted returns by 9.8 basis points, with a t -statistic of 2.39. In column 2, with a year-by-month fixed effect, this increases slightly to 11.6 basis points. In the specifications including year-by-month fixed effects, the top quarter of days experienced returns 7.1 basis points higher (column 4, with a t -statistic of 4.27), and the top third of days experienced returns 5.7 basis points higher (column 6, with a t -statistic of 3.71). Column 8 shows that days with above-median dividend payments experience returns 4.7 basis points higher, with a t -statistic of 3.15. The constant indicates that below-average days earn returns of 1.7 basis points. This means that above-median days earned returns of 6.4 basis points, nearly four times higher than the below-median days.

An alternative way to quantify the magnitude of the effect is to ask how much of the equity premium is earned on days with high dividend payments. We explore this question in Table 3. The first row shows the average annual return to investing on days with high dividend payments—the “Top” columns—and the returns to investing on days with low dividend payments—the “Not” columns. Column 1 shows that investing in the top 2 weeks earns an average annual return of 1.2 percent, while investing on all other days earns a return of 6.8 percent. This means that the top 2 weeks per year (3.8 percent of trading days) earn about 16 percent of the equity premium. This ratio of 4.06 indicates that the top 2 weeks account for roughly quadruple the amount of the equity premium compared to the number of days they are in the market. The final two columns examine being above or below the median dividend payment. The above-median days experience average annual returns of 6.0 percent, while the below-median days experience returns of 1.7 percent. This indicates that the above-median days experience roughly 78 percent of the equity premium, even though they only account for 50 percent of the days.

TABLE 3—EQUITY PREMIUM BASED ON DIVIDEND PAYMENT

	2 weeks		Quarter		Third		Median	
	Top	Not	Top	Not	Top	Not	Top	Not
Avg. annual return	0.012	0.068	0.049	0.028	0.052	0.026	0.060	0.017
Percent of equity premium	15.5	84.5	63.4	36.6	67.1	32.9	77.7	22.3
Percent of days	3.8	96.2	24.8	75.2	33.2	66.8	50.0	50.0
Percent premium/percent days	4.06	0.88	2.55	0.49	2.02	0.49	1.55	0.45

Notes: This table shows how the annual market return and equity premium varies with the dividend payment yield. Columns labeled “Top” show values when the column header dummy variable is equal to 1, and rows labeled “Not” show values when the dummy variable is equal to 0. The first row shows the average cumulative return for each calendar year. The next row shows the fraction of the total (the sum of the Top and Not first row) the average in the first row represents. The third row shows the fraction of days each dummy variable is equal to 1. The final row shows how much of the equity premium was earned compared to the number of days the dummy variable was equal to 1 (i.e., row 2 divided by row 3). “2 weeks” is a dummy variable equal to 1 if the payment yield is in the top 10 of the prior 252 trading days. “Quarter” is a dummy variable equal to 1 if the payment yield is in the top 63 of the prior 252 trading days, “Third” is a dummy variable equal to 1 if the payment yield is in the top 84 of the prior 252 trading days, and “Median” is a dummy variable equal to 1 if the payment yield is in the top 126 of the prior 252 trading days.

Next, we consider how the above results can map into a trading strategy. While there are many possible ways to operationalize the predictability of market returns from dividends, they share the same essential aspect of changes in market exposure depending on the current dividend payment or, equivalently, changes in leverage. Thus, a strategy can be described as a rule for choosing leverage on day t applied to the excess return of the market, namely, $Leverage_t \times (Market_t - Rf_t)$.

Our goal is not to develop an optimal strategy that maximizes ex post performance but rather to show transparent and simple strategies that illustrate the magnitude of the effects that we document.²¹ Thus, similar to Figure 1, we base our strategy on the quintile of a day’s abnormal dividend relative to the distribution over the prior year (so that the number can be calculated ex ante). In order to avoid having performance reflect a *net* increase in average leverage over time, we design strategies that have an average leverage of approximately one (i.e., the same as just buying and holding the market). With this in mind, a binding constraint to a strategy is the maximum leverage that can be used. Our sample period is long, and the frictions involved with obtaining leverage likely exhibit variation over time. Thus, we present strategies with different maximum values for leverage. Our lowest leverage strategy has a maximum leverage of 150 percent, which we call Max 1.5x; our Max 2x strategy sets a maximum leverage of 2, while our Max 3x strategy has a maximum of 3. In modern markets, these levels are easily attainable even by retail investors, and at least anecdotally, sophisticated arbitrageurs have utilized similar levels of leverage for a long period of time.²² For the Max 1.5x strategy, the top quintile of dividend

²¹Optimizing this strategy would take into account more nuanced variation in the relationship between returns and dividends, in addition to issues such as trading or tax-related costs, which both vary over our long time series. For example, the specifics of an optimal strategy would be different for implementation by a hedge fund versus within an active ETF wrapper or for a retail investor, not to mention how such a strategy was overlaid with additional strategies. With that said, the essential underlying strategy is likely to be similar to what we describe below.

²²For example, levered ETFs currently provide levered exposure to the market at levels greater than 3x. 1.5x leverage is considerably under the maximum values for margin accounts established under Regulation T, whereby an investor can borrow up to 50 percent of the purchase price of securities. So by depositing cash, an investor could

TABLE 4—MARKET RETURNS BASED ON DIVIDEND PAYMENT: ALTERNATIVE DIVIDEND NORMALIZATIONS

	No div payment				Div payment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mkt div pay[$t - 1, t$]	57.39 (3.16)	65.12 (3.34)	70.20 (3.46)	74.39 (3.52)	40.83 (2.06)	45.91 (1.94)	24.27 (1.20)	16.09 (0.62)
YM fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Value weight	Yes	Yes	No	No	Yes	Yes	No	No
Equal weight	No	No	Yes	Yes	No	No	Yes	Yes
R^2	0.000644	0.0494	0.000942	0.0855	0.000142	0.0486	0.0000404	0.0601
Observations	24,537	24,537	24,537	24,537	24,355	24,355	24,355	24,355

Notes: This table shows how the market return varies with the dividend payment yield. The return is restricted to stocks that did not pay a dividend on days t and $t - 1$ in the first four columns. The return is restricted to stocks that did pay a dividend on either t or $t - 1$ in the last four columns. Even-numbered columns contain year-by-month fixed effects. t -statistics based on heteroskedasticity-robust standard errors are in parentheses.

payment days lever the market at 1.5, the fourth quintile levers at 1.25, the third quintile at 1, the second at 0.75, and the bottom at 0.5. For the Max 2x strategy, the top quintile has a leverage of 2, with a decrease of 0.5 for each quintile so that the bottom quintile is completely out of the market (a leverage of 0).²³ Finally, for the Max 3x strategy, the top quintile has a leverage of 3, the fourth quintile a leverage of 2, and the bottom three quintiles are out of the market (a leverage of 0).

We present the excess returns to these strategies in Table 5. The unlevered excess market return is 2.8 basis points per day. The Max 1.5x strategy earns 3.4 basis points, and the Max 2x strategy earns 4.0 basis points, while the Max 3x strategy earns 5.2 basis points. While the standard deviations are slightly higher for the levered strategies, this increase in volatility is less than the increase in mean, such that the levered strategies produce higher Sharpe Ratios: 0.030, 0.032, and 0.033, respectively, relative to 0.026 for the market (i.e., increases of 15 percent, 23 percent, and 27 percent, respectively). Daily market returns are not a particularly common unit, so the table also lists annualized returns. The market in our sample had an annualized excess return of 7.2 percent. The Max 1.5x strategy earned 8.7 percent, the Max 2x leverage strategy earned 10.6 percent, and the Max 3x strategy earned 14.0 percent. Figure 2 shows the cumulative returns to a \$1 investment in 1926. By 2018, \$1 invested in the excess return in the market was worth \$210, while our strategies had values of \$812, \$2,548, and \$20,293, respectively.

Finally, to show that the overall returns are not just due to the returns from the largest days, in the Supplemental Appendix we run the same strategy ignoring the 10 largest predicted days of returns each year. Consistent with the results in Tables 2 and 3, these days are substantial drivers of returns, and dropping them results in a 15 percent decrease in the market excess return to 0.024. With that said, we find considerably higher returns and Sharpe ratios to the strategies in this sample.

in principle buy up to twice that amount in equities under Regulation T and achieve 2x leverage. 1.5x leverage allows for a further 50 percent haircut on collateral value if the collateral were held in other equities, even without considering the additional market exposure or beta that would be provided by the collateral in such a case.

²³Results are similar if leverage is continuously changed based on percentile values over the same leverage ranges, rather than using quintile breakpoints.

TABLE 5—TRADING STRATEGIES

	Market-Rf (1)	Max 1.5x (2)	Max 2x (3)	Max 3x (4)		
<i>Panel A. Trading strategy</i>						
Mean [% daily]	0.028 (4.01)	0.034 (4.65)	0.040 (4.96)	0.052 (5.18)		
Mean [% annual]	7.197	8.869	10.567	14.045		
Std dev	1.067	1.123	1.245	1.560		
Sharpe ratio	0.026	0.030	0.032	0.033		
Mkt adjusted return[%]		0.007 (3.02)	0.014 (3.02)	0.029 (3.28)		
α [%]: CAPM		0.006 (2.74)	0.013 (2.74)	0.029 (3.26)		
α [%]: 3-factor		0.006 (2.50)	0.011 (2.50)	0.028 (3.15)		
α [%]: 3-factor + momentum		0.005 (2.26)	0.011 (2.26)	0.026 (2.93)		
Avg leverage		1.009	1.019	0.995		
Market beta		1.018	1.036	0.983		
Observations	24,015	24,015	24,015	24,015		
		Max 1.5x	Max 2x	Max 3x		
		Q1 Q4	Q1 Q4	Q1 Q4		
<i>Panel B. Trading strategy by quarter</i>						
Sharpe ratio	0.043	0.036	0.048	0.035	0.060	0.024
α [%]: CAPM	0.012 (2.97)	0.005 (0.99)	0.025 (2.97)	0.011 (0.99)	0.065 (4.04)	0.008 (0.41)
α [%]: 3-factor	0.011 (2.57)	0.006 (1.09)	0.023 (2.57)	0.012 (1.09)	0.064 (3.69)	0.010 (0.49)
α [%]: 3-factor + momentum	0.011 (2.57)	0.004 (0.86)	0.023 (2.57)	0.009 (0.86)	0.064 (3.69)	0.005 (0.24)
Avg leverage	1.011	1.011	1.021	1.021	1.004	1.004
Market beta	1.062	0.995	1.124	0.990	1.141	0.938
Observations	5,923	6,066	5,923	6,066	5,923	6,066

Notes: This table shows the performance of portfolios from various trading strategies. All strategies take the returns to a strategy long the CRSP VW market and short the risk free, $(Market - Rf)$, combined with leverage, namely, $Leverage \times (Market - Rf)$. In panel A, the first column shows the returns to $(Market - Rf)$ unlevered (a constant leverage of 1). The second column shows our Max 1.5x strategy, which uses leverage of (0.5, 0.75, 1, 1.25, 1.5) for each quintile from 1 to 5, respectively. Column 3 shows our Max 2x strategy, which utilizes leverage of (0, 0.5, 1, 1.5, 2) for each quintile. Column 4 shows our Max 3x strategy, which uses leverage of (0, 0, 0, 2, 3) for each quintile. Leverage is based on the rank of the abnormal dividend payment (the sum of dividends paid on days t and $t - 1$ divided by the average daily dividend payment on days $t - 20$ through $t - 272$) on the given day compared to the prior 252 days. The first row shows the average of each portfolio, with the t -statistic for the test that this equals zero underneath in parentheses. The following row shows the annualized version of the first row, $(1 + return)^{252} - 1$; the next row shows the standard deviation; and the following row shows the Sharpe ratio. The Mkt Adjusted Return [%] row shows the strategy return minus the market excess return, with the t -statistic underneath. The α [%]:CAPM row shows the α (in percent) from the regression of the strategy on the excess return of the market, with the t -statistic underneath. The α [%]:3 - Factor row conducts a similar analysis adding the size factor (SMB) and the value factor (HML) (Fama and French 1993). The α [%]:3 - Factor + Momentum adds the momentum factor (UMD) (Carhart 1997). The following row shows the average of leverage, and the next row shows the beta coefficient from regressing the daily strategy return on the market excess return. Panel B shows statistics based on data only from the first quarter (odd columns) or the fourth quarter (even columns).

Of course, beyond means and volatilities, a strategy's value largely lies in its alpha, which is a measure of its worth above and beyond included return

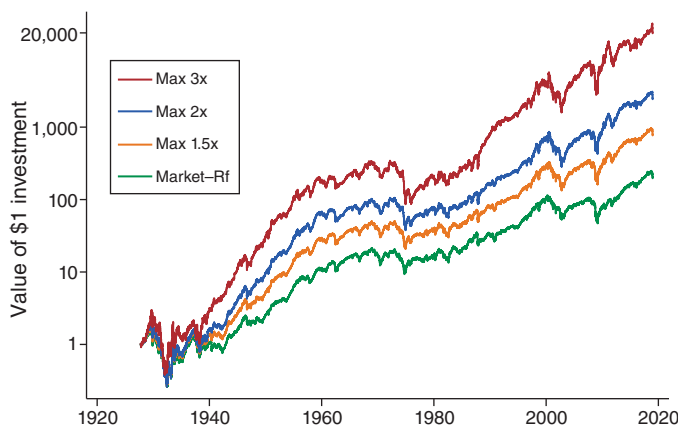


FIGURE 2. CUMULATIVE RETURN TO \$1 INVESTMENT BASED ON VARIOUS TRADING STRATEGIES

Notes: This graph shows the cumulative return to strategies that invest \$1 in January 1926 through 2018. The green line shows the excess return of the value-weighted market portfolio. The other three strategies utilize strategies based on the percentile of the rank of the current days dividend, $\text{Mkt Div Pay}[t - 1, t]$ relative to that variable over the prior year (252 trading days). In the orange line the daily market return is levered for each of the five quintiles at levels of $[0.5, 0.75, 1, 1.25, 1.5]$. In the blue line the daily market return is levered for the five quintiles at levels of $[0, 0.5, 1, 1.5, 2]$. For the maroon line, the five quintiles are levered at levels of $[0, 0, 0, 2, 3]$. The y-axis is presented in a log scale.

predictors.²⁴ Table 5 shows the alphas of the strategy relative to the CAPM, a 3-factor model (Fama and French 1993), and a 3-factor model with the addition of momentum (Carhart 1997). The alphas are each economically large and statistically significant. The alpha of the Max 3x strategy is of a similar magnitude to the average daily return of the market. The alphas represent the expected return above and beyond a strategy hedging out the beta adjusted factors on the right-hand side of the regression. The fact that the alphas are each similar to the simple difference between the strategy return and the market excess return demonstrates that the strategy is largely independent of standard cross-sectional return predictors, which is consistent with it being a pure time series strategy. A related result in Table 5 is that the average daily market beta of each strategy is about one. This shows that while the long-only strategy is exposed to market risk, and by necessity has more exposure to market risk on particular days, in the time series average, it is not more exposed than a simple buy-and-hold investor. This can be seen in Figure 2 by the fact that the lines generally move together while fanning out over time. This intuitively makes sense, as markets move for many reasons, and most of these movements are not explained by dividend payments.

Of course, arbitrageurs need not focus on a single strategy (and in practice, they tend not to), which means that there is no need for an arbitrageur to allocate capital to a strategy consistently throughout the year. Section IVE shows that returns are larger at the beginning of the year and are largely absent at the end, likely related to the time-varying demand for cash to make end-of-year payouts. Thus, we repeat our

²⁴ We note that this is an unconditional alpha. The conditional alpha is of course zero since the strategy is just leveraging up and down the market portfolio. We thank an anonymous referee for this clarification.

analysis looking separately at the first and fourth quarter of the year. In the fourth quarter, the alphas are not statistically significant, meaning the strategy does not outperform holding the market. The first quarter displays a very different pattern. Examining the alphas, the Max 1.5x strategy outperforms the market by more than 1.1 basis points per day, the Max 2x by 2.3 basis points, and the Max 3x strategy by more than 6.4 basis points. In the first quarter the triple-levered strategy earns 10 basis points per day (an annualized return of 29 percent), with a Sharpe ratio of 0.06, more than double that of the market. For an arbitrageur following multiple trading signals, they could choose to allocate more capital to the dividend payment strategy in this early portion of the year.

Section VB discusses the implication, interpretation, and tradability of these strategies, and aspects of this interpretation vary based on different assumptions of what the market understood *ex ante*.

B. Returns between Paying and Nonpaying Firms and the Impact of Other News Events

A question that naturally arises is whether the effects represent price pressure in the companies that paid the dividend or elsewhere in the market. Hartzmark and Solomon (2019) document that mutual funds and institutions rarely reinvest dividends into the stocks that they came from. If this is the case, price pressure should not be limited to the firms that actually paid the dividend.²⁵

In Table 4, we consider how the effects are different for the firms paying dividends that day versus other firms and find the strongest effect for firms that do not pay the dividend. When examining non-dividend-payers (columns 1–4), the results are similar in magnitude and significance to Table 1. When only firms paying dividends on those days are included, the results are smaller—around two-thirds as large for value-weighted returns (columns 5 and 6) and one-quarter to one-third as large, and statistically insignificant, for equal-weighted returns (columns 7 and 8).

A potential concern is that dividend payments vary with other predictors of daily market returns. Returns vary over the week (with the lowest returns on Monday and the highest on Friday), as well as over the month (with the highest returns at the turn of the month (Lakonishok and Smidt 1988), as well as with macroeconomic announcements (e.g., Lucca and Moench 2015; Savor and Wilson 2013). Table 6 shows after controlling for each of these, the coefficient on dividend yield remains positive and significant, suggesting that these patterns do not account for our result. Meanwhile, the inclusion of year-by-month fixed effects controls for longer-horizon time series predictors of returns, which are generally only measured at lower frequencies, typically quarterly or annually (Goyal, Welch, and Zafirov 2024).

We also examine the effect of dividend and earnings news. Because the dividend news was announced long before the payment date, under standard theories of market efficiency, there will be no response to any information from the announcement

²⁵ It is difficult to predict the relative magnitudes of effects for paying and nonpaying firms. Even if the majority of money is invested outside of the firms that paid, there are many more nonpaying firms. Thus, we predict *some* price pressure for nonpaying firms, but the relative amount of price pressure for the two groups is ultimately an empirical question.

TABLE 6—MARKET RETURNS BASED ON DIVIDEND PAYMENT: CONTROLLING FOR DAILY EVENTS

	Day of week	Macro announce		Div announce			Earn surprise	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mkt div pay[$t - 1, t$]	55.71 (3.43)	251.1 (2.67)	264.3 (2.80)	64.08 (2.82)	68.47 (2.99)	64.46 (2.35)	129.1 (2.61)	128.0 (2.59)
Div announce[$t - 1, t$]					62.34 (1.44)			
Div announce news						−8.819 (−0.17)		
Earning surprise								15.35 (3.13)
Event control	Yes	No	Yes	No	Yes	Yes	No	Yes
YM fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0565	0.0332	0.0364	0.0460	0.0461	0.0711	0.0376	0.0388
Observations	24,537	6,294	6,294	14,348	14,347	10,228	7,817	7,817

Notes: This table shows how the market return varies with the dividend payment yield controlling for patterns in returns related to calendar events, macroeconomic announcements, dividend announcements, and earnings surprises. The first column includes dummy variables for the day of the week as well as a turn-of-the-month dummy variable equal to 1 if it is the last day of the month or the first three days of the month. Our next tests are based on variables that are not available for the full sample, which limits the time frame upon which we can conduct our analysis. We examine FOMC and macro announcements from 1994, dividend announcements from 1962, and earnings surprises from 1984. To facilitate comparison with specifications that include controls, we rerun our baseline specification without controls (but covering the same sample periods as the control variables are available) in columns 2, 4 and 7. Column 3 includes dummy variables for FOMC announcements, CPI, PPI, employment, and GDP announcements. Column 5 controls for the market level of dividend announcements calculated the same as Mkt Div Pay[$t - 1, t$] substituting dividends announced as the numerator (instead of dividends paid in our baseline variable). Next, we include dividend news, which is the amount of dividends announced on day t and $t - 1$ minus the amount announced on the same two calendar dates one year prior, scaled by market capitalization. Column 8 controls for earnings surprise calculated as the surprise of large firms announcing on day t , as in Hartzmark and Shue (2018). Market return is measured in percent as the CRSP value-weighted market index. Market payment yield is measured as the cumulative payment yield on day t and $t - 1$ (Market Div Pay[$t - 1, t$]). All columns contain year-by-month fixed effects. t -statistics based on heteroskedasticity-robust standard errors are in parentheses.

on the payment date. However, it is possible that payment amounts coincide with some other news, such as information contained in that day's dividend announcements or information in that day's earnings. We consider multiple versions of these, including dividend amounts announced that day, dividend news announced, and the earnings surprise announced.²⁶

The results in Table 6 show that neither dividend amounts nor dividend news significantly explain market returns, even on the day when the economic content of the news is first released. By contrast, earnings announcements *do* significantly drive market returns. However, crucially for our results, including any of these measures leaves the effect of dividend payments substantially unchanged.

²⁶ Dividend amounts are calculated analogously to our payment yield measure but using announced amounts, namely, the sum of dividend amounts announced on t and $t - 1$ scaled by total market capitalization. Dividend news is the difference between the dividends announced on t and $t - 1$ minus the dividends announced on those dates one calendar year prior, scaled by market cap (see Section IVC for a discussion of this measure). Finally, we consider earnings surprises, as in Hartzmark and Shue (2018). For each firm, we measure the difference between announced earnings and the median analyst forecast made over 15 days before the announcement. We then take the top 10 percent of firms by market capitalization, and value weight their surprise.

The insignificant effect of dividend announcements illustrates that the market-wide news from most dividend announcements in practice tends to be minor. This is because dividends are typically paid in persistent nominal amounts according to a rule chosen specifically because firms do not want to deviate from it (e.g., Brav et al. 2005; Baker, Mendel, and Wurgler 2016). Thus, prior to a dividend announcement, the date it will occur and the amount that will be announced are generally straightforward to predict (e.g., Hartzmark and Solomon 2013; Bessembinder and Zhang 2015). This stickiness of dividends means that the surprise component of a dividend announcement is likely to be much smaller than, and quite different from, the quantity announced.

This result also speaks to whether the returns we document would be a good use of arbitrage capital. In practice, when sophisticated investors implement a market timing strategy, the goal is not to identify pure-play strategies but rather to layer strategies onto one another to form the best prediction of returns from all possible predictors. Thus, an important question is whether dividend-based predictability has additional predictive power above and beyond known predictors because if it does, market timing strategies are likely to find it useful. The fact that predicted dividend payouts continue to explain returns *ex post* after controlling for these other factors (and, indeed, the coefficients remain stable) suggests that it would be a valuable addition.²⁷ In the Supplemental Appendix, we consider another way of showing the marginal usefulness of dividends: computing strategy alphas on days when other market timing signals do not give predictions, such as days without an FOMC announcement and days of the week when returns are generally low (Mondays and Tuesdays). We find that implementing the dividend strategies on those days only continues to have positive and significant alphas.

In the Supplemental Appendix we also show the results of a simple strategy combining market-level predictability based on dividend payments and FOMC announcements. Perhaps the strongest predictor of day-to-day market returns is the FOMC announcement premium, which Lucca and Moench (2015) shows accounts for nearly all of the equity premium.²⁸ Layering the FOMC announcement premium into our dividend strategy increases the alpha on our Max 2x strategy from about 1 basis point to over 2 basis points, while in the Max 3x strategy, the alpha increases from about 3 basis points to about 4 basis points. The alpha to this strategy is significantly above the market excess return of about 3 basis points. The average return to the combined Max 3x strategy over both dividends and FOMC announcements is more than 7 basis points per day, which is an annualized return of about 20 percent.

²⁷In principle, a market timing strategy could also be layered on top of a cross-sectional strategy (e.g., buying value stocks in general but using dividend payouts to adjust the leverage). While it is less straightforward to know exactly how these interactions would work, our results are suggestive that the explanation for our predictability does not obviously lie in the fact that arbitrage capital would view it as a worse substitute for their existing strategies. Further, the vast majority of cross-sectional strategies involve portfolio sorts at monthly horizons or longer. Given our results are not greatly impacted by including year-by-month fixed effects, our predictability is largely unrelated to these anomalies.

²⁸In earlier data the day-of-the-week effect was strong, but this predictability is largely absent in recent data and so would be unlikely to be the basis for a recent strategy (see Supplemental Appendix).

C. *Delayed Reactions*

The previous results already test for the possibility that dividend payment date amounts are proxying for economic news revealed that day, either from dividend, earnings, or macro announcements. However, an alternative possibility is that the payment date amounts represent a delayed reaction to *prior* days' economic news. The most obvious of these would be if the payment dates involve a delayed reaction to news in the dividends being paid out that day. In such a scenario, there would be a link between the payment amounts and daily returns, but due to a delayed reaction to information rather than price pressure.

The first hurdle for such an alternative is that dividend news does not appear to be important for the market. This is seen in the lack of a significant effect for dividend announcements in the previous table. Theories of delayed reaction generally posit that the news is important but not fully responded to immediately. In the case of earnings, for instance, the large contemporaneous response to earnings news is insufficient, leading to post-earnings announcement drift (e.g., Bernard and Thomas 1989 and many others). In the current instance, it would need to be the case that the dividend news is relevant on the payment date despite *not* being relevant on the announcement date. In other words, investors must miss all of the economic content of the news until the later date (to explain why there are significant effects on payment dates but not announcement dates).

When considering such a scenario, it is important to distinguish between the dividend *amount* being announced and the dividend *news* being announced. The amount is the quantity of dollars that an investor will receive. The news is the *surprise* component, the difference between what the investor expected to be announced and what was actually announced. In theories of information transmission, whether rational or behavioral, the intuition is the same going back to at least Samuelson (1965); it is the surprise component that moves prices. The part that was expected prior to the announcement will not cause prices to move. Given the stickiness of dividends in practice discussed above, dividend amounts are likely quite different from measures of dividend news.

In addition, theories of a delayed response to this surprise component as an explanation for payment day returns have to overcome several difficulties. First, the gap between the announcement and payment averages over 40 days, so this cannot be a matter of a delay of a few days.²⁹ More importantly, any delayed reaction to news from the announcement must not occur with a general lag but on the payment date specifically. In other words, even if people do not pay full attention when dividends are announced, they must be jolted out of their inattention specifically on the day of payment. It seems *ex ante* unlikely that this is due to news focusing on the firm itself, as both dividend announcement dates and ex-dividend dates generally receive more

²⁹One potential concern is that there could be exceptions that are important in predicting returns where the time is too short for the market to respond before payment. Such instances are rare, as the first percentile of the time between announcement and payment is 17 days, and the twenty-fifth percentile is 29 days. The Supplemental Appendix shows similar results to our baseline restricting the sample to observations where a significant period of time elapses between the announcement and the payment.

attention for firms than payment dates.³⁰ A more complex mechanism could be that the payment itself triggers a recollection of the dividend news, whereby investors react as if it was new information. This also seems unlikely, as Hartzmark and Solomon (2019) show that most brokerage platforms do not announce dividend payments or link them clearly to the firms that paid them. Such recollection-triggering explanations are further complicated by the fact that the news would need to apply to firms other than the one that paid the dividend, as the results above show that this is where the returns are concentrated and that the news does not seem that economically relevant in the first place

Nonetheless, we wish to test whether a delayed reaction could explain our results. In order to do this, we would like a variable that is related to the amount being paid (so it predicts reinvestment amounts) but which is plausibly unrelated to most versions of the surprise in the recent announcements. To do so, we predict dividend payments using data from long before the most recent dividend announcement. Since dividends are sticky in timing and amount, past years' dividend amounts are likely a reasonable proxy for current amounts. To proxy for the current day's dividend yield, we use the dividend yield from a prior year on the same calendar date. We examine lags between 1 and 10 years. For example, with a one-year lag, if we are examining returns on March 25, 1982, we use the dividend payment yield from March 25, 1981. We do so all the way to 10 years, which predicts returns from March 25, 1982, using the payment yield from March 25, 1972.

When we use dividend predictions from many years prior, we are able to cleanly separate the dividend amount from the dividend news. The 10-years-prior amount is a decent forecast for the current amount, but it is hard to see why it should forecast the recent surprise. Even if investors take several months to fully process information (i.e., surprise), as in Coibion and Gorodnichenko (2015) or other frameworks with delayed reactions to news, it is unlikely that they are still reacting to the specific news a decade later, or that the 10-years-ago amount is related to surprise in some other news between announcement and payment, or anything else. Indeed, because of the difference between news and amounts, it is not even clear that the 10-years-ago amount is even a strong proxy for 10-years-ago news, much less recent news. While it is difficult to rule out every possible mistake, very stale amounts are unlikely to be correlated with most versions of current surprise—certainly full-information rational expectations versions but also most behavioral versions of delayed response to recent information.

Before examining whether the measure of prior years' dividends predicts returns, it is worth asking if such a crude proxy is a meaningful predictor of the current year's dividend payment. To do so, we regress the current dividend payment yield from year y on the lagged payment yield from the same date in year $y-L$. To capture any longer-term predictability, we also include a year-by-month fixed effect. We repeat this exercise using a 1-year lag and continue to a 10-year lag.

Table 7, panel A shows that this simple rule yields significant predictability. Examining the 1-year lag in column 1 gives a coefficient of about 0.8, with a

³⁰For example, the *Wall Street Journal* currently includes a "Dividend Changes" list in the market statistics it presents. This list includes firms that recently announced their dividend and does not include firms when they pay the dividend.

TABLE 7—MARKET RETURNS BASED ON LAGGED DIVIDEND PAYMENT

	1 yr lag (1)	2 yr lag (2)	3 yr lag (3)	4 yr lag (4)	5 yr lag (5)	6 yr lag (6)	7 yr lag (7)	8 yr lag (8)	9 yr lag (9)	10 yr lag (10)
<i>Panel A. Current dividends predicted by dividends in prior years</i>										
Mkt div pay	0.789 (210.45)	0.646 (116.03)	0.640 (100.31)	0.757 (119.40)	0.822 (193.88)	0.758 (198.36)	0.582 (113.91)	0.525 (84.23)	0.576 (88.18)	0.679 (121.34)
Within R^2	0.719	0.483	0.417	0.501	0.670	0.672	0.474	0.350	0.373	0.498
R^2	0.756	0.558	0.513	0.581	0.716	0.717	0.553	0.457	0.486	0.580
Observations	18,449	15,485	15,149	15,273	19,597	20,249	15,446	14,210	14,073	15,824
<i>Panel B. Current returns predicted by dividends in prior years</i>										
Mkt div pay	69.69 (3.64)	62.79 (3.45)	88.79 (5.02)	51.29 (2.73)	34.47 (2.06)	73.16 (4.49)	55.83 (3.54)	73.58 (4.87)	81.95 (5.42)	58.85 (4.19)
Observations	18,449	15,485	15,149	15,273	19,597	20,249	15,446	14,210	14,073	15,824
<i>Panel C. Current returns predicted by current dividends restricting sample to panel B observations</i>										
Mkt div pay	92.24 (3.79)	59.58 (2.51)	90.76 (3.41)	53.23 (2.20)	54.73 (2.51)	94.76 (4.52)	69.73 (2.60)	80.44 (4.53)	85.68 (4.88)	59.51 (3.68)
Observations	18,449	15,485	15,149	15,273	19,597	20,249	15,446	14,210	14,073	15,824

Notes: This table shows how the current day's dividend is predicted by dividends from that day in prior years (panel A) and how market returns vary with prior years' dividends (panel B). Each column uses dividend payments of the indicated lag. For example, if the left-hand-side variable is from day D , month M , and year Y , then the column labeled " X Yr Lag" indicates the dividend payment yield matched to that observation is from day D , M , and year $Y - X$. Panel C performs the baseline regressions of contemporaneous measures of dividend payment on returns on the sample of observation included for a given lag in panel B. All regressions contain year-by-month fixed effects. Panel A reports the within R^2 , which is the R^2 after accounting for the influence of the fixed effects. t -statistics based on heteroskedasticity-robust standard errors are in parentheses.

t -statistic of over 200. In terms of evaluating the predictability, though, the important statistic is the within R^2 , which indicates how much of the variation this lagged value predicts after removing the influence of the fixed effects. We find that this simple rule using data that are a full year stale yields an adjusted R^2 of 0.719. Given this simple proxy is able to account for such a large fraction of the day-to-day variability in the dividend payment, we feel confident using it as a daily dividend proxy in the analysis that follows.³¹

Given the strong predictability of the lagged dividend measure, in panel B we regress returns on day t in year y on the two-day dividend payment yield from day t year $y-L$, where L is the number of years lagged indicated by each column. Examining column 1, we see a coefficient of 69.69, with a t -statistic of 3.64. Days without a payment yield, which occur if the day on the prior year occurred on a day without trading, such as a weekend, are dropped. Thus, panel C conducts our baseline analysis on the subsample of days with dividend information for comparison to the results in panel B. For a 1-year lag, the sample loses about 6,000 days, and on

³¹ We note that this is an intentionally simple rule that could be improved upon in a variety of dimensions. It likely would be possible to go to the company level and better predict patterns, such as firms that prefer certain days of the week, or that correct for weekends, or any number of improvements. Simply lagging the date and not correcting for weekends means we only match about 70 percent of trading days. The goal of this analysis is to show how predictable these values are using the most simple rule possible. Thus, the coarse measure here offers the best illustration of the predictability of these flows, rather than a better, but significantly more complex measure.

this subsample our baseline regression has a coefficient of 92.24, with a t -statistic of 3.79. Given we are using a noisy proxy of daily dividend payment, we would expect some attenuation bias. Nine of the 10 point estimates in Table 7, panel B are lower than the point estimate in panel C, consistent with this prediction, although many of the differences are quite small. We find strong results using data as much as 10 years old. With a 10-year lag, we see a point estimate of 58.85, with a t -statistic of 4.19. This suggests that these dividends are highly predictable and that our dividend variables likely are capturing the influence of the dividend amount, not a reaction to the dividend news.

As an alternative way to show that news in the dividend announcement is unlikely to be driving payment day returns, in the Supplemental Appendix we also explicitly control on the payment date for proxies for the level of dividend news relative to the same payment date in years prior. We have shown above that the dividend paid on the same date a calendar year prior is a strong predictor of the current date. Thus, a reasonable proxy of the payment date surprise on a given date should be captured by $news_{(M,D,Y)} = Dividends_{(M,D,Y)} - Dividends_{(M,D,Y-1)}$, which is the difference in dividends on a given day relative to the prior year.³² Even though this “surprise” aspect is already preannounced, it may still drive returns under a delayed reaction version. However, this payment day surprise variable has little explanatory power over returns. The coefficient on news has a t -statistic of 0.15 without fixed effects and 0.28 with fixed effects included. Adding the news controls leaves the coefficient on dividend amount unchanged.

The results above rule out a number of explanations based on a delayed reaction to information, but it is difficult to rule out all variations. The hardest to preclude are those that include both a delayed response to the initial announcement that occurs on the payment date, combined with a biased perception of the announced news (in order to make announcements with minimal news have a significant influence on prices). For example, investors could forget until the payment date that dividends are paid *at all* and only remember on the payment date, thereby treating the entire dividend amount as dividend news. In this instance, even the 10-year stale amount of dividends will be correlated with the current perceived surprise. Such a version would still be interesting from the perspective of both behavioral finance, and short-term market predictability, although the mechanism driving the returns would be different from what we describe in this paper. While such explanations may predict the base result, they have more difficulty explaining the other patterns we document, such as why the effect should be greater at the start of the year than the end, why it should vary with liquidity, or why it should be increasing over time. Each of these tests are natural predictions of price pressure but are not nearly as obvious if the cause is a delayed response to news that occurs with a bias.

³²This is a rough proxy but seems likely to capture significant variation in such news as long as an investor updates to information over a time period less than a year. We acknowledge that actual responses to news need not match this linear form. With these caveats, it seems that if delayed reaction to such news accounts for our results, this proxy should capture a significant portion of it and at least be directionally correct.

TABLE 8—INTERNATIONAL MARKET RETURNS BASED ON DIVIDEND PAYMENT

	(1)	(2)	(3)	(4)
Mkt div pay[$t - 1, t$]	20.30 (2.19)	30.21 (3.59)	22.65 (3.02)	26.89 (4.50)
Country YM FE	No	Yes	No	Yes
Date	No	No	Yes	Yes
R^2	0.0000330	0.0611	0.268	0.309
Observations	237,185	237,070	236,775	236,660

Notes: This table shows how market returns vary based on the dividend payment yield across 58 international markets. Data are in a panel format, with data from a market/day combination included when the market reports information for at least 100 stocks. The market return is calculated as the value-weighted averages of those stocks. Market payment yield is measured as the cumulative payment yield on day t and $t - 1$ in that market. Columns 2 and 4 include month-by-market fixed effects. Columns 3 and 4 include date fixed effects. Standard errors are clustered by date and market, with t -statistics in parentheses.

D. International Evidence on Dividend Payments and Market Returns

Table 8 switches to an international setting and considers how daily dividend payments predict market returns across 58 international markets.³³ These tests examine predictable price pressure while abstracting away from a number of possible institutional details that are unique to the United States. We again focus on the dividend payment on day t and $t - 1$. While the payment mechanisms will vary from country to country, there is a reasonable assumption that payment dates and/or the day after should roughly correspond to when investors can use the payment to purchase securities. To the extent that institutional details lead to mismeasurement of reinvestment, or that the data are less accurate, we expect this to bias against finding a result.

In Table 8, column 1, with no controls, we find that a higher dividend payment yield is associated with higher returns, with a coefficient of 20.30 and a t -statistic of 2.19. Column 2 adds country-by-year-by-month fixed effects, and the coefficient increases to 30.21 with a t -statistic of 3.59. The economic magnitudes of these effects are somewhat smaller than in the United States. A 1 standard deviation increase in payment (0.000389) predicts higher returns in column 2 by 1.2 basis points.

This panel setting means that we can compare returns across countries on the same day based on the difference in countries' dividend payments. Column 3 adds a day fixed effect (that is to say, controlling for all global news that day) and finds similar magnitudes to column 1, but with a larger t -statistic. Column 4 includes date

³³These 58 markets are the subset with sufficient data from the 125 markets in Compustat Global. A given market and day are included when it contains at least 100 stocks with prices. The countries in the analysis are Australia, Austria, Belgium, Bangladesh, Bulgaria, Bermuda, Brazil, the Cayman Islands, Chile, China, Croatia, Cyprus, the Czech Republic, Denmark, Egypt, Finland, France, Great Britain, Germany, Greece, Guernsey, Hong Kong, India, Indonesia, Ireland, Israel, Italy, Japan, Jordan, Korea, Kuwait, Luxembourg, Malaysia, Mexico, the Netherlands, Nigeria, New Zealand, Norway, Pakistan, Peru, the Philippines, Poland, Romania, Russia, Saudi Arabia, Singapore, Spain, Sri Lanka, South Africa, Sweden, Switzerland, Thailand, Turkey, Taiwan, Ukraine, United Arab Emirates, and Vietnam. Data cover 1986–2017.

and country-by-year-by-month fixed effects, and the significance increases again, to a t -statistic of 4.50.

E. Returns Based on Reinvestment Intensity

Prior research and results in Section IVH suggest that retail investors rarely reinvest dividends. Thus, it is likely that the returns we document are largely related to the reinvestment rate of professional investors. If a fund is about to send cash to investors and receives a dividend from its holdings, it may not be worth it to pay the transaction costs to hold the new shares only for a short time. As a consequence, reinvestment rates are likely to be lower as the fund's own dividend payment draws near. In the United States, funds and ETFs are subject to the pass-through rule, whereby to avoid paying corporate income tax at the fund or ETF level, all dividends and realized capital gains (minus fund expenses) must be distributed to the fund's investors by the end of the year. This leads many funds to wait until the end of the year to pay out most of their dividends and realized capital gains. The Supplemental Appendix shows the average proportion of fund annual payouts distributed each quarter. Consistent with the intuition above, the graphs indicate that a significant fraction of mutual fund payouts, nearly two-thirds, occur during the fourth quarter. The next panel graphs the payouts by month and shows that much of this pattern is driven by December payouts, which constitute roughly 60 percent of the annual payout.³⁴

If money managers are less likely to reinvest when they are about to send cash to investors, and if they are responsible for a significant portion of the price effects, then the effects should diminish prior to these large payouts. In other words, a given dollar of dividends paid will lead to less reinvestment and less price pressure. Given the pattern in dividend payments, we would expect the lowest reinvestment prior to the large December payouts. This suggests the lowest price effects in the fourth quarter and December in particular. On the other hand, once the next year arrives, given the long time until the December payouts, we expect the most reinvestment at the beginning of the year.

Table 9 explores whether this is the case. It repeats our baseline analysis but examines how the coefficient on dividend yield varies by quarter and includes a year-by-month fixed effect to ensure the regressions are not picking up variation in the level of returns in a given calendar month. Column 1 shows that the magnitudes of the coefficients are consistent with the price pressure predictions. The weakest effect is in the fourth quarter when the largest payouts occur. We find insignificant coefficients that are close to zero during this time. Also consistent with the predictions, the largest effect is during the first quarter, which is the farthest away from the large end-of-year payments. While the graph of mutual fund payments suggests

³⁴ Our analysis here is based on the timing of ex-dates for these payments, as we lack the actual payment dates for mutual funds. Nonetheless, the gap between ex-dates and payment dates appears shorter than for stocks, making it likely that the monthly or quarterly timing of ex-dates maps to payment dates. For example, in recent years, this gap has been five to six days for the Fidelity Magellan fund, two days for the Vanguard Total Stock Market Index Fund, and zero to three days for the iShares S&P 500 Index Fund. We use all fund distributions (dividends and realized capital gains), since they require cash to be paid, regardless of the label for that cash, but the patterns for dividend payments alone are similar.

TABLE 9—MARKET RETURNS BASED ON DIVIDEND PAYMENT BY QUARTER AND MONTH

	Value weighted		Equal weighted	
	(1)	(2)	(3)	(4)
Q1 × mkt div pay	134.8 (2.40)		228.8 (3.66)	
Q2 × mkt div pay	54.92 (1.68)		31.60 (0.92)	
Q3 × mkt div pay	73.78 (2.44)		53.16 (1.91)	
Q4 × mkt div pay	12.01 (0.35)		−6.124 (−0.17)	
Not December × mkt div pay		82.91 (3.89)		89.99 (3.96)
December × mkt div pay		−48.92 (−1.25)		−56.87 (−1.30)
YM fixed effects	Yes	Yes	Yes	Yes
R^2	0.0501	0.0501	0.0872	0.0862
Observations	24,537	24,537	24,537	24,537

Notes: This table shows how the market return varies with the dividend payment yield based on the calendar quarter or month. Market return is regressed on Market Div Pay $[t - 1, t]$ interacted with time period dummy variables. Columns 1 and 3 include quarter-of-year dummy variables, and columns 2 and 4 include dummy variables for whether the calendar month is December or other months of the year. All columns contain year-by-month fixed effects. *t*-statistics based on heteroskedasticity-robust standard errors are in parentheses.

a strong quarterly pattern, the monthly graph suggests that much of this pattern is specifically due to December. To test this, we split the effects of dividend yield into December versus all other months. Consistent with this, we find no significant effect of dividend payment on market returns in December.

In addition to patterns within the year based on payout frequency, there have been large time series shifts in the popularity of investment vehicles like ETFs and mutual funds. Such products seem particularly likely to reinvest dividends, so one might expect the patterns we observe to become more pronounced as these products become more popular. Figure 3 suggests a significant increase in the effect coinciding with the rise of mutual funds and ETFs. The figure shows regression coefficients of the market return on the dividend payment yield on day t and $t - 1$, conducted separately each decade. For example, the point graphed for 1970 is the regression coefficient using daily data from 1970 to 1979. Prior to the 1990s, the dividend payment coefficient was always positive, with a mean of about 50 and a maximum of 83. The three decades post-1990 represent the three largest coefficients in our sample, each greater than 100. These results further support the interpretation of the main results being due to price pressure, as the size of the effect from a dollar of dividends varies with the reinvestment rate.³⁵ Meanwhile, it also confirms the

³⁵ Interestingly, while the coefficients are larger in recent periods, the standard errors are also larger. While we lack a particular prediction for this, it is consistent with the idea that when funds and ETFs become larger, they have a larger effect on average, but flows into and out of those funds for other reasons can also lead to larger swings that may obscure the effects of reinvestment. This is also consistent with the existing evidence that funds' reinvestment choices are significantly driven by their own cash flow requirements.

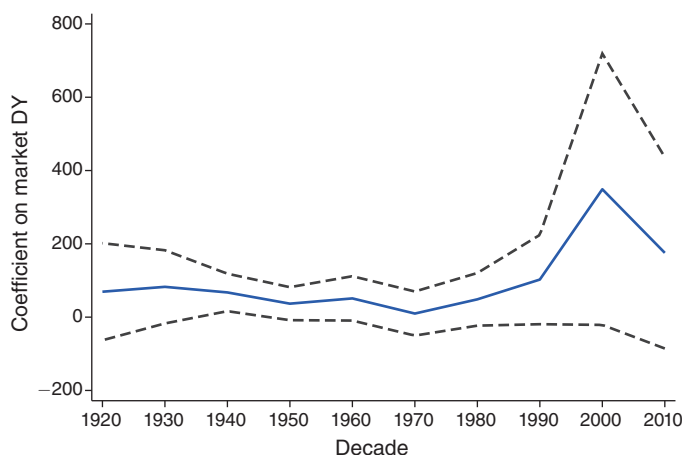


FIGURE 3. RETURN SENSITIVITY TO DIVIDEND PAYMENT BY DECADE

Notes: This graph shows the coefficient of value-weighted market returns regressed on Market Div Pay $[t - 1, t]$ conducted separately for each decade. Each data point represents a decade; for example, the data point for 1930 is the regression coefficient when the regression is conducted on all data occurring from 1930 through 1939.

importance of financial intermediaries like mutual funds, who are responsible for much of the actual dividend reinvestment in the market.

F. Returns Based on Market Liquidity

An additional prediction of price pressure is that its impact should be larger when there is less liquidity. Similar to Nagel (2012), we proxy for liquidity using measures of market volatility based on the VIX index, which uses implied volatility from S&P 500 options.³⁶ The VIX data we examine begin in 1990, so we also use the News VIX index developed in Manela and Moreira (2017), which is available at the beginning of our sample. VIX is a noisy proxy for market-wide liquidity that likely captures other market dynamics. To demonstrate robustness using an alternative, we also construct a market-wide measure of daily Amihud illiquidity (Amihud 2002).³⁷ While market-wide liquidity is an elusive concept that is difficult to measure, we think that showing the results using two separate measures, each with their own shortcomings, should assuage concerns that idiosyncrasies of the measure drive the result. With that said, given the difficulty of concretely measuring market-wide liquidity, we think the results should be taken as suggestive but not conclusive.

Table 10 examines how the impact of dividend payments interacts with levels of the VIX and News VIX in predicting market returns. We augment our baseline regression by including VIX and also an interaction of VIX with the dividend payment yield.

³⁶Theoretically, the VIX index should be strongly correlated with the returns to liquidity provision (e.g., Nagel 2012; Gromb and Vayanos 2002; Brunnermeier and Pedersen 2009; Adrian and Shin 2010; Ang, Gorovyy, and Van Inwegen 2011; Ben-David, Franzoni, and Moussawi 2011).

³⁷We follow the steps in Amihud (2002) and Amihud and Mendelson (2015) in constructing the measure. Specifically, we take the measure at the individual stock level, drop the highest 1 percent of observations per day, and value weight the remaining observations. We multiply the measure by 10^7 .

TABLE 10—MARKET RETURNS BASED ON DIVIDEND PAYMENT AND ILLIQUIDITY MEASURES

	VIX		News VIX		Amihud	
	(1)	(2)	(3)	(4)	(5)	(6)
ILLIQ $\times$ Mkt div pay	30.08 (3.79)	24.12 (2.95)	10.74 (2.89)	11.53 (3.05)	17.72 (5.88)	18.29 (5.79)
Mkt div pay	-416.5 (-2.47)	-240.6 (-1.40)	-210.1 (-2.22)	-227.0 (-2.36)	0.250 (0.01)	0.579 (0.03)
YM fixed effects	No	Yes	No	No	No	Yes
Year fixed effects	No	No	No	Yes	No	No
R^2	0.0191	0.122	0.00198	0.00690	0.00355	0.0562
Observations	7,304	7,304	23,822	23,822	24,537	24,537

Notes: This table shows how the market return varies with the dividend payment yield interacted with measures of liquidity. Liquidity (ILLIQ) is proxied for by the VIX (columns 1–2), the News VIX (columns 3–4), and Amihud illiquidity from day t (columns 5–6). Value-weighted market returns are regressed on three separate variables: the illiquidity measure alone (not reported), Market Div Pay $[t - 1, t]$, and the interaction of the two. Columns 2 and 6 include year-by-month fixed effects, and column 4 includes year fixed effects. t -statistics based on heteroskedasticity-robust standard errors are in parentheses.

Column 1 examines the coefficient on this interaction term and shows a significant coefficient of about 30. Column 2 shows a similar coefficient of about 24 after controlling for year-by-month fixed effects. A 1 standard deviation increase in the VIX of 7.8 leads to an increase in the sensitivity of dividend payment of 187. This suggests that in times when VIX is higher, the impact of dividend reinvestment is also higher.

A potential concern with these results is that they only cover the period from 1990 onward and thus may be driven by a relatively short sample. In columns 3 and 4, we use the News VIX index to explore the entire sample. All of the major patterns are the same. The 11.5 coefficient in column 4 suggests that a 1 standard deviation increase in the News VIX of 4.6 coincides with an increased sensitivity of dividend payment of about 53. This is about 80 percent of our baseline coefficient of 67.

Columns 5 and 6 find similar results using Amihud illiquidity from day t . The coefficient on the interaction term of 18 implies that a 1 standard deviation increase in illiquidity of 3.1 is associated with a higher coefficient on dividend payment of about 56. This is nearly identical to the estimate using the News VIX. These results further support the interpretation of price pressure, with each dollar of dividends having more effect in periods when there is less liquidity.

G. Long Horizon Returns

Returns around dividend payment dates offer sharp tests of the price pressure prediction because of the clean variation provided by daily payments. Nonetheless, these may not reflect the full extent of price pressure, as some investors may reinvest dividends with a delay. Longer horizons also test whether these returns quickly reverse afterward.

While the effects are not as tightly identified from timing, we consider how daily payment yields predict returns at horizons out to a quarter. To do so, we regress cumulative returns at different horizons on a dummy for whether day zero was

above the median dividend payment, plus a constant.³⁸ We conduct this regression separately for each horizon, beginning with just the payment date alone and extending to the cumulative return from the payment date to 60 trading days after the payment date, and graph the coefficient on the dummy variable. Thus, each point can be interpreted as how much higher the cumulative return was by that point after a day with above-median payments. We run versions of this regression with no controls as well as versions including year fixed effects to control for time trends and daily future dividend payments (during the period of the cumulative return) to ensure we are not capturing price pressure from future payments.

Figure 4, panel A graphs the results without controls, and panel B includes controls. Each point is the regression coefficient on the payment yield using the cumulative return from $t = 0$ to the x-axis value. The first week shows similar results of strong price pressure of about 10 basis points. After this, the pattern is noisier. With that said, the effect increases over roughly the first month, peaking around $t + 20$. Taking the point estimates seriously suggests that reinvestment continues at a lower rate in weeks two through four. This may arise if some investors are inattentive about the timing of dividend payments (consistent with the observation in Hartzmark and Solomon (2022) that dividend payments are often not flagged in brokerage platforms). The noisiness of these results makes it difficult to distinguish between additional buying pressure or a lack of meaningful returns.

Perhaps most interestingly, the results show no evidence of a short-term reversal in the first month, even in point estimates. Before seeing these results, a plausible hypothesis might be that the price pressure we observe quickly reverses, having minimal impact on any long-term measure. None of the regressions present evidence consistent with a short-term reversal.

Extending further in the future, the noise increases substantially. By the end of roughly a quarter (60 days), we are unable to reject either the hypothesis that the effect has totally reversed or the hypothesis that it is permanent. Our point estimates suggest that this is not an effect that quickly reverses. Unfortunately, we lack the statistical power to measure whether the effect has a permanent impact or reverses over longer horizons. While cross-sectional reversals have been studied in a variety of contexts, much less is known about the speed and extent one should expect entire markets to reverse price pressure. We think this is an important avenue of study for future research.

H. Price Multipliers

The previous analysis of dividend payments and market returns considered economic magnitudes primarily in terms of the observed variation in the dependent variable—that is, how much larger are market returns with higher dividends. Another important metric of economic magnitude is the multiplier of market price with respect to a dollar of predictable investment. This is the inverse of the price elasticity (the change in quantity demanded as a function of price changes). In order to estimate such a figure, we need an estimate of what fraction of a dollar of

³⁸ We examine abnormal dividend yield to define the dummy variable so as to not induce a positive relationship between scaled price measures and longer period returns in the specifications without controls.

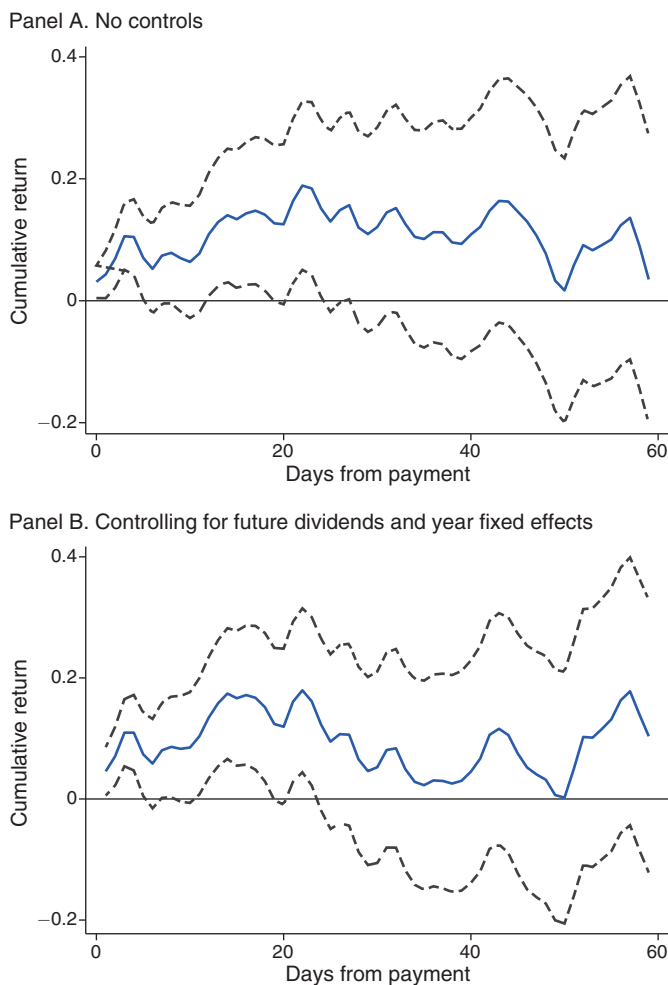


FIGURE 4. LONGER-TERM RETURNS BASED ON DIVIDEND PAYMENT

Notes: These graphs show the difference over time between returns on days with an above-median dividend payment and returns on days with a below-median dividend payment. Specifically, the cumulative return from $t = 0$ through the number of days indicated on the x-axis is regressed on a dummy variable equal to 1 if the day is above the median dividend payment and a constant. This coefficient is graphed in blue, and its 95 percent confidence interval is shown by the gray dashed line. The second graph includes controls for year fixed effects, as well as future dividend payments included up to the number of days on the x-axis.

dividends is typically reinvested into the market. That is to say, we need to approximate the number of dollars spent on purchases, not the number of dollars received. While data limit the ability to provide exact figures, this section provides a range of such estimates and what they imply for the market's demand elasticity.³⁹

One potential (though implausible) benchmark is to assume that all funds are reinvested on either the payment date or the following day. Under this assumption, the coefficient of 67 from Table 1 implies a multiplier of 0.67. This suggests that

³⁹ An important potential improvement for future work would be to have more accurate data to allow for more precise estimates of predicted trades.

TABLE 11—PRICE MULTIPLIER ESTIMATES

Multiplier	Aggregate reinvestment %	Retail (34%)	Mutual funds and ETFs (28%)	Other (38%)
0.67	100	100	100	100
1.02	66	0	100	100
1.27	53	0	53	100
1.92	35	0	53	53

Notes: This table presents estimates of price multipliers under different assumptions about reinvestment rates. Estimates are based on the coefficient of 67.07 from Table 1, panel A, column 4. The first column presents multiplier estimates based on the assumption for aggregate reinvestment rate in column 2. The multiplier estimate is 67.07 divided by the aggregate reinvestment percentage. The aggregate reinvestment rate is based on the assumptions about reinvestment rates for retail investors (column 3), mutual funds and ETFs (column 4), and all other investors (column 5). The aggregate reinvestment % implied by these numbers is the average of columns 3 through 5, weighted by the proportion of each investor type in the market, which is listed in the column headings and is based on the March 2020 Financial Accounts of the United States release Z.1.

a \$1 reinvestment leads to an increase in market value of \$0.67. Table 11 lists the implied multiplier based on the reinvestment rate, with the first row showing the estimate of 0.67 assuming that all investors invest all dividends.

The estimate of 0.67 is an implausible lower bound for the multiplier but is already too high to be explained by any standard asset pricing model; 100 percent reinvestment is intuitively unlikely and inconsistent with the behavior of some investors. In order to estimate aggregate reinvestment, we take the fraction of aggregate holdings from the Financial Accounts of the United States release Z.1 from March 2020. We then estimate reinvestment rates for these investor types. The Supplemental Appendix discusses these estimates in detail. We estimate that retail investors (about 34 percent of the market) do not meaningfully quickly reinvest dividends. We estimate that mutual funds and other professional investors reinvest 53 percent of dividends. These estimates yield an aggregate reinvestment rate of 35 percent and a multiplier of 1.92.

V. Discussion

A. Price Pressure, Limit Orders, and Supply and Demand

This paper explores shocks that are relatively small individually but which occur nearly every day (and so cumulatively are large). This stands in contrast to most of the literature, which explores large shocks that occur infrequently. We argue that small and frequently occurring shocks, while being less dramatic, are in fact highly important for understanding price pressure more generally.

The logic of how and why price pressure occurs is often made significantly more confusing by the common practice of trying to understand price movements through the lens of a vertical supply and a horizontal demand curve. When explaining microeconomics to undergraduates, market participants are partitioned into consumers of a product who make up the downward-sloping demand curve and producers who make up the upward-sloping supply curve. The price is given by the intersection

of those curves, so an increase in buyers (an increase in demand) or a reduction in sellers (a decrease in supply) leads to a price increase. In the case of stocks, the most common modeling assumption is rather different. Supply is usually described as the number of shares outstanding, fixed in the short term. This means that the supply curve is a vertical line (see Shleifer 1986) for a discussion of the standard intuition). The (net) demand curve, of buyers minus sellers, is assumed to be horizontal.

This formulation yields intuition about market dynamics from a representative agent under the efficient market hypothesis, namely, that the market will bear a net demand of any quantity at the prevailing price, with quantity fixed by the number of shares. But it thereby obscures the effect of different traders' beliefs or liquidity needs. The day-to-day trading of stocks is not represented by interactions of supply and demand but occurs *within* the net demand curve. This framework muddies the question of whether price pressure is even a coherent concept. Not infrequently, one hears objections to the effect that "one investor is buying, one investor is selling, and the quantity of shares is fixed. How can investors in aggregate have price pressure? Pressure on what?"

This confusion seems to stem more from the textbook model, not the plausibility of the price pressure concept. For example, standard terms like "a shift in demand" are unclear for horizontal demand. For downward-sloping demand, it makes no difference if the demand curve is considered to be transposed *vertically* (i.e., a willingness to pay a higher price for each quantity) or *horizontally* (i.e., a willingness to buy a greater quantity at each price) because the final effect is the same. Indeed, the expressions "demand shifts up," "demand shifts to the right," and "demand shifts up and to the right" are generally used as *synonyms*, meaning that many economists likely have not considered which version they have in mind.

For horizontal demand, this equivalence is removed. An increase in quantity demanded given the price (a horizontal transposition) has no influence. It leads to an identical line, as unconstrained net demand is infinite at the market price and is zero at a price of one penny more. However, if a shift in demand is interpreted as a shift in the willingness to pay (a vertical transposition), this leads to a shift in price. These two different predictions lack a distinct vocabulary, making it challenging to grasp how the model is meant to work.

An alternative conception, and we feel a more useful one, is to consider demand and supply in terms of people bringing trades to market.⁴⁰ In such a scenario, supply is no longer the number of shares outstanding. Rather, it is the number of shares of a stock that would arrive as sell orders over the horizon of interest if the price shifted.⁴¹ Similarly, demand is the number of shares sought in buy orders under the same circumstances.

⁴⁰ At a specific moment in time, this maps closely to the concept of the limit order book (aggregated over all trading venues). For most economic applications, the instantaneous elasticity is not the length of time over which price impact is evaluated, so it is worth extending the concept over the relevant horizon, such as a day or a month, etc. There is no real-world analog to this, but the theoretical goal is to capture investors in the current limit order book along with all those who would enter the market when prices shift.

⁴¹ We feel that this formulation yields significantly more intuition, as the number of shares outstanding is not generally a constraining factor in the number of sell limit orders that can be placed. Short sellers can generally borrow shares and generate greater numbers of sell limit orders. Even the company itself can also be reduced to a potential seller or purchaser of shares rather than the main determinant of supply.

When a sufficiently large liquidity-demanding order is placed, it will eat through the limit orders in the book and eventually will affect the last trade price. The question is what happens next. If aggregate net demand is perfectly elastic in the standard framework, this is equivalent to saying that this theoretical limit order book over some period of time has *infinite depth at both the bid and the ask*. Put this way, this assumption is far from innocuous. At short horizons we can observe a finite depth of the current limit order book for all securities.

Thus, to motivate the perfectly elastic demand assumed by standard models, there must be an arbitrage mechanism functioning beyond the current limit order book. The standard assumption is that investors will realize that the price has deviated from fundamentals and place large trades in the opposite direction to restore the price to its previous level. While this could be the case, perhaps they won't, either due to lacking capital; or thinking the trade isn't worth it; or being uncertain whether the counterparty knows something they don't; or trading in the opposite direction, thereby exacerbating rather than correcting the mispricing (e.g., Brunnermeier and Nagel 2004). In any case, the idea of offsetting limit orders to replace price-affecting trades (even if economically intuitive in certain contexts) is an additional assumption layered on top of the mechanics of the limit order book. In other words, price pressure is a general and largely *mechanical* prediction, whereas offsetting trades are an economic prediction arising from particular models of how investors trade.

Thus, when considering the influence of trade on prices, the first-order question is which traders are providing liquidity to offset orders and under what circumstances. In this regard, the existing evidence from large and infrequent shocks tends to focus on situations where it is unclear, if not unlikely, that there actually will be substantial numbers of arbitrageurs trying to offset the price movements. Due to the low frequency of these shocks, it is hard for this to comprise a dedicated fund strategy, and the small number of events makes diversification difficult. Further, large shocks necessitate a large amount of capital to trade against them, which either requires many traders or very large bets from a smaller number of traders.⁴² Finally, some of the results (e.g., fire sales) are focused on periods where financial markets are stretched and thus working in nonstandard ways.

These concerns likely contribute to the revealed preference of many asset pricing papers, which is to use models where trade in and of itself has no influence on prices. The evidence of price pressure in extreme settings, where arbitrageurs may struggle to offset price movements (or not even attempt to do so), need not be relevant to understanding price movements in normal times.

We note that, before seeing the results in our paper, this is a perfectly coherent framework upon which to view asset markets. While it is unclear who provides liquidity against large and infrequent shifts in demand, small and predictable shocks are exactly what liquidity providers are assumed to be in the business of offsetting. For example, the NYSE has designated market makers whose job is to "take the other side of trades when buying and selling imbalances occur."⁴³ Further, these liquidity providers are assumed to be deep-pocketed enough to offset such normal size flows. A key piece of evidence for deciding whether trade

⁴²Gabaix and Koijen (2021) note that in practice, there is limited arbitrage capital at the macro level.

⁴³<https://www.investopedia.com/terms/d/designated-market-maker-dmm.asp>

ought to be a central component of understanding price movements is what happens with small and predictable shocks, when liquidity providers should be best able and most likely to offset price movements. Our paper provides such evidence and finds that such flows influence prices. Dividend payments represent perhaps the most easily identified source of normally recurring flows, but in the course of the market, there are myriad other examples that are more difficult to identify. This suggests that small, normally recurring flows likely generally lead to shifts in prices.

B. Interpretation of Results under Various Assumptions

While our results illustrate the importance of price pressure, some implications of the findings depend on questions that we are not able to directly answer. We discuss these below and how they affect the understanding of our findings.

Are the Shocks Recognized as Uninformed?—An interesting question is whether potential arbitrageurs were aware of the pattern that we document. Given that dividend payments are obvious and recurring, the standard market efficiency prediction is that the pattern ought to be understood by informed investors. If these trades are understood to be uninformed, standard asset pricing models suggest they should not influence prices.

However, our results suggest that if arbitrageurs know about the pattern, they are unwilling or unable to correct it. One potential explanation could be that this deviation from market prices is not significant enough for arbitrageurs to view it as profitable (e.g., Grossman and Stiglitz 1980). The time series magnitudes and associated trading strategies that we document are relatively sizable, though the costs from implementing them may also be considerable. The costs would be less of a concern if (as would be likely in practice) the strategy were used as an overlay to an existing strategy rather than as a stand-alone strategy. If costs prevent the strategy from being acted upon by arbitrageurs, even as an additional piece of a strategy, it suggests that the range of mispricing they can correct may be more limited than imagined and that a key component of arbitrageur behavior is absent from standard models.

While arbitrageurs perhaps *ought* to know about the findings in this paper, it is not clear that they actually *do*. The implicit assumption in the idea that informed traders know about a newly documented mispricing pattern is that academics are the last to know about it. This assumption is far from innocuous and appears at odds with the empirical evidence that anomalies tend to decline partially after they are published (e.g., McLean and Pontiff (2016) estimate a 58 percent decline after publication), with the degree of decline strongly correlated to their tradability. Mundane daily predictable basis point fluctuations may be the type of setting that people focus less attention on, making it less likely that arbitrageurs engage in offsetting trades, leading to price changes unrelated to information flows (similar to Da, Guren, and Warachka 2014 and Chang et al. 2016).

With that said, the space of possible flows with price impact is large, and many of them are likely to be more difficult to understand or forecast than the ones we study. Thus, if investors are unaware of this pattern, this suggests that liquidity provision by arbitrageurs is insufficient to offset price impact from other, less obvious flows.

This would imply such liquidity provision is present only in certain obvious situations and is not the ubiquitous force typically assumed.⁴⁴

One interesting test of the tradability of a documented price pattern is what happens after the paper documenting it becomes widely known. If the effects were understood prior to the paper but arbitrageurs did not wish to correct them, one would expect such effects to persist. If the effects were not understood but the knowledge becomes widespread and induces trading, we would expect the pattern to change out of sample, consistent with the alpha decay in McLean and Pontiff (2016).

While only time will tell with respect to dividend predictability and market returns, we can examine the out-of-sample performance of a similar market-level return pattern that became widely known almost 40 years ago, namely, day-of-the-week anomalies. While some findings were known prior, Lakonishok and Smidt (1988) systematically demonstrated the importance of day of the week in predicting market returns over a long time period, which received significant attention both within industry and academia. Further, the magnitude is roughly comparable to what we explore. Above-median days of the week have returns of about 5 basis points, with other days closer to 0, which is similar to the roughly 4 basis points spread we find splitting our dates at the median.⁴⁵ Consistent with market-wide anomalies of this magnitude being tradable, if we repeat the Lakonishok and Smidt (1988) exercise post-1989 (when the idea was widely understood), we find much less predictability of returns based on days of the week (see the Supplemental Appendix). This suggests that a market-wide predictor of a similar magnitude to what we find that would be exploited with a similar strategy to how one would exploit dividend predictability was viewed as a profitable-enough trading opportunity for arbitrageurs to trade against. It also suggests that anomalies can persist (for about a century) when the main limit to arbitrage is not an economic friction but rather the lack of knowledge of the opportunity.

Finally, we note that if arbitrageurs notice such mispricing, their incentive may not be to trade to correct the mispricing (the “sunshine trading” idea of Admati and Pfleiderer 1991) but rather to front-run it (the “predatory trading” idea of Brunnermeier and Pedersen 2005).⁴⁶ These different possibilities mean that not every method of acting on the mispricing will reduce it in the short term.

Whether these return patterns were understood *ex ante* or only became known after this paper suggests somewhat different underlying mechanisms accounting for the results. Either of these mechanisms are consistent with the importance of price pressure in a way that is incompatible with standard models, so directly testing these channels is an important avenue of future research.

⁴⁴We note that if the pattern we document disappears due to arbitrageur trading after the publication of the paper, this strengthens this point. It suggests the market missed this obvious pattern, corrected it once it became aware of it, and likely continues to miss other, more difficult-to-identify patterns.

⁴⁵Based on Table 2 of Lakonishok and Smidt (1988). We note both our strategy and the day of the week have further nuance than just the median split (e.g., holidays or large dividend days).

⁴⁶Bessembinder et al. (2016) argue that in the case of ETFs predictably rolling futures contracts over, the evidence is more consistent with sunshine trading.

Permanent or Transitory Shocks?—Next is the question of how long the effect persists in market prices. As discussed above, our results can effectively rule out a short-term reversal occurring over a few days. Ex ante we feel that many would have predicted that such an effect would quickly reverse, thus demonstrating that the effect is present after multiple weeks is interesting.

Our results looking at longer horizons are noisy. Beyond about a month, our effect is not statistically different from zero, and the confidence interval expands looking out to a quarter. With that said, at a quarter our effect is also not statistically different from the magnitude after a month. Thus, our results are consistent with an effect that either reverses over roughly a month to a quarter or an effect that persists beyond such a time frame. While reversals have been studied at length in the cross section, it is unclear how such results relate to the time series. Thus, it is difficult to know at what speed and to what degree a pattern like this should reverse at the market level.

Intuitively, many financial economists think reversals must occur, which is the prediction of almost every asset pricing model written before about 2020. However, the arguments that imply that price pressure must reverse quickly are similar to the arguments that price pressure should not exist in the first place. As a result, evidence against the latter set of assumptions ought to call into question the former ones as well. Gabaix and Koijen (2021) build a model where neither aspect is true—price pressure exists and has permanent effects. They list the expectation of a flow-based reversal as a “tenet of finance” that changes if their inelastic market hypothesis is correct. Empirically, they face similar limitations to our paper whereby at long horizons their confidence intervals expand and the degree of reversal cannot be estimated sharply. Understanding whether or not there is a reversal, and its magnitude for the market in response to flow-based price changes, is an important empirical question in the ongoing debate about modeling asset prices that is left for future research.

Finally, our paper studies shocks that are small, common, and operating over short horizons. This distinguishes it from shocks that are large, common, and operate over long horizons (e.g., fund flows in Gabaix and Koijen 2021) as well as flows that are large, uncommon, and short term (e.g., index reconstitutions). Inferring how the market will react to one class of shock from the other is not entirely straightforward, given that the background question is “How much, and under what circumstances, do arbitrageurs supply liquidity?” Our results provide insight into an important class of shock that has not been examined before. This complements other studies of different kinds of shocks, as it is not clear that there is a single answer for how liquidity provision works in all circumstances.

VI. Conclusion

In this paper, we show that predictable increases in the number of people buying or selling shares lead to predictable changes in price for those shares. Put in such stark terms, such a finding may seem obvious. And yet, many of the standard models in finance assume this result does not hold—that when trading stocks, investors are effectively exchanging 20-dollar bills for 100-dollar bills, and the price for such an exchange will be constant regardless of volume. Even when trades are considered to move prices, the general assumption is that the main force is information about the

trade, not the trade itself. We consider price changes in the largest and most liquid assets (the value-weighted market portfolio), where the flows in question ought to be known to not result from private information and where the timing and amount of such flows is known in advance. Even with these aspects stacking the deck against the possibility of price pressure, price pressure is evident.

One interesting question is whether one would expect price pressure to be more pronounced at the individual stock level or at the market level. Academic finance generally predicts that the larger and more liquid the asset, the more likely that the asset behaves in accordance with the efficient markets hypothesis. To the extent that value-weighted market returns are driven considerably by the returns of the largest stocks, and these tend to have stronger measures of liquidity (smaller spreads, greater depth and volume, more information disclosure) compared with most other stocks, one might expect the market to thus be more resilient to uninformed trade. It is unclear if this is the appropriate null hypothesis, though. The alternative view is that while individual stocks may be close substitutes for each other, there are relatively few good substitutes for the market as a whole, an argument made by Samuelson (1965) (as reported in Jung and Shiller 2005). One way of interpreting our results is that, for whatever reason, investors have strong views as to which assets they want to hold, and changes in the price of such stocks do not greatly change their willingness to purchase them. The question of *why* this is the case is one of great interest.

Finally, our results suggest a reevaluation of the predictability of market returns at short horizons. Shiller (2014), when summarizing the literature on return predictability in his Nobel Prize address, states that not only *is* there not much predictability at short horizons empirically but that there *should not* be as a theoretical matter (citing Sims 1984). As he describes,

It is easy to see why short-term forecastability of price changes in investable assets should in some sense be unlikely: if investment returns were substantially forecastable from day to day, it would be too easy to get rich in a year or so by trading on these forecasts, and we know it cannot be easy to make a lot of money trading. ..., the low R^2 in many tests of short-run market efficiency are neither surprising nor interesting. The tests tell us only the obvious, and do not tell us about the rationality of markets ... (Shiller 2014, pp. 1490–91)

Our results suggest that the relationship between the different versions of predictability—“low R^2 values,” “statistically significant predictors,” and “potentially profitable trading strategies”—is more nuanced than might be inferred from the quote above. Consistent with the Shiller intuition, our R^2 values are very low. Despite this, we find that there are statistically significant predictive variables and that they can generate economically important returns and trading strategies that are profitable before expenses. The low R^2 is indeed obvious along one dimension: Daily market returns exhibit enormous variation, and most of it is not due to dividend payments. Nonetheless, it would be a mistake to conclude from this that there are no variables of economic significance that can predict returns in a manner likely to be of interest to academics or practitioners. The question of how many other short-term predictors

there might be, and what they reveal about the underlying forces driving price movements, is one of considerable interest.

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